MASTER OF BUSINESS ADMINISTRATION

SEMESTER 1



DMBA117

DATA VISUALIZATION

Unit – 9

Creating Interactive Dashboards in Tableau

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1. INTRODUCTION

The era of big data has transformed how organizations analyze and interpret information, making dashboards an essential tool for data-driven decision-making. Dashboards serve as a dynamic interface for exploring, visualizing, and understanding complex datasets. Chapter 8 focuses on the step-by-step process of creating these dashboards, starting with the fundamentals of building standalone charts and gradually integrating them into cohesive, interactive dashboards that bring clarity and insight to data.

To ensure an engaging and meaningful user experience, this chapter explores into techniques for combining multiple visualizations into a single dashboard. Readers will learn how to create seamless connections between charts, allowing for fluid navigation and in-depth analysis. Interactivity, a key feature of modern dashboards, is emphasized through the addition of filters, enabling users to focus on specific data subsets that align with their exploration goals. By empowering users to control what they see, filters make dashboards more personal and impactful.

Finally, the chapter explores advanced dashboard actions such as highlights and interactivity enhancements, enabling users to uncover hidden patterns and relationships within their data. By the end of this chapter, readers will have the tools and knowledge to design dashboards that not only communicate information effectively but also engage users in an immersive data exploration experience. Whether you're an analyst, developer, or business professional, these techniques will empower you to unlock the full potential of your data.

1.1. Objectives

After studying this unit, you should be able to:



- Learn how to construct visually appealing and functional dashboards that enable efficient analysis and insights.
- Master the techniques for integrating various charts into a unified interface to present data comprehensively.
- Understand how to add and configure filters, allowing users to explore specific subsets of data dynamically.

- Discover how to use highlighting techniques to draw attention to significant trends and patterns in dashboards.
- Apply advanced interactive features, such as linking charts and triggering actions, to create immersive and insightful dashboards.

2. INTERACTIVE DASHBOARDS - DEFINITION

Interactive dashboards in Tableau are powerful tools that allow users to visualize, explore, and analyze data dynamically. Unlike static reports, Tableau dashboards enable interactivity, providing users with the ability to interact directly with the data through elements such as filters, highlights, and tooltips. These dashboards combine multiple visualizations, charts, and metrics into a single interface, offering a comprehensive view of the dataset while allowing users to drill down into specific details as needed.

The interactive features in Tableau dashboards enhance user engagement by making data exploration intuitive and customizable. Filters, for instance, enable users to focus on specific subsets of data, while actions such as highlights and cross-filtering allow seamless navigation between related data points across visualizations. This interactivity transforms dashboards from being merely informational to becoming tools for decision-making, enabling stakeholders to uncover insights that may otherwise remain hidden in static representations.

Furthermore, Tableau's user-friendly interface empowers both technical and non-technical users to create and interact with dashboards. Features like drag-and-drop functionality and customizable layouts ensure that dashboards are not only visually appealing but also tailored to the specific needs of the audience. Whether used for operational monitoring, strategic planning, or storytelling, interactive dashboards in Tableau are essential for organizations aiming to leverage data for actionable insights in real time.

3. BUILDING DASHBOARD FOR DATA EXPLORATION

Dashboards in Tableau allow users to combine multiple visualizations into a single interactive workspace, enabling comprehensive data exploration and analysis. Here's a step-by-step guide to building dashboards and enhancing their interactivity:

Step 1: Creating a New Dashboard

After creating individual sheets (visualizations) in Tableau, the next step is to combine them into a dashboard. Creating a dashboard is straightforward and follows a process similar to creating a new worksheet:

- Access the Dashboard Workspace:
- Locate the tabs at the bottom of your Tableau workbook.
- Click the New Dashboard icon (represented by a rectangle with a "+" symbol). This opens a blank dashboard workspace.



Step 2: Adding Views to Your Dashboard

Once the dashboard workspace is open, the next step is to populate it with visualizations from your existing sheets. Tableau makes it easy to incorporate these views and organize them to create an effective, user-friendly dashboard. Here's how to add and manage views:

Accessing the Sheets List

1. On the left panel of the dashboard workspace, you'll find the Sheets list. This panel displays all the visualizations (sheets) you've created in your workbook.
2. Identify the views you want to include in your dashboard. Each sheet represents a specific data visualization, such as a bar chart, line graph, or map.

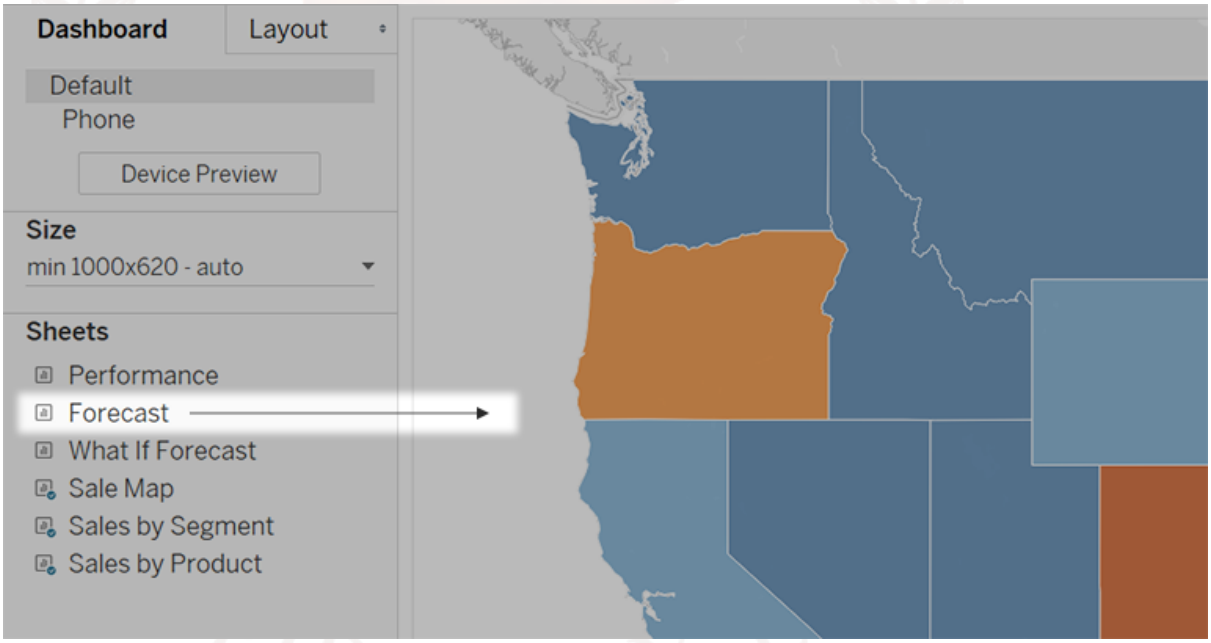
Dragging and Dropping Views

1. **Select a Sheet:**
   - Click and hold the name of the sheet in the Sheets list.
2. **Drag the Sheet to the Dashboard Workspace:**
   - Move the sheet from the Sheets list and drop it onto the dashboard workspace on the right.

- Tableau will automatically add the sheet as a tile in the layout.

Arranging the Views

1. **Positioning the View:**
   - Drag the view to the desired position within the dashboard. Tableau's grid layout makes it easy to align elements.
   - Use the tiled layout to snap views into fixed positions, or switch to the floating layout for more flexibility in placing elements.
2. **Resizing the View:**
   - Adjust the size of the view by clicking and dragging its edges. This ensures that the view fits well within the overall dashboard design.



Step 3: Replacing a Sheet in the Dashboard

In Tableau, you can easily replace an existing sheet in a dashboard without redoing the layout or reapplying settings like filters, borders, and padding. This feature is especially useful when you need to update or swap a visualization to provide a different perspective or correct an error. Here's how to do it:

Selecting the Sheet to Replace

1. **Identify the Sheet to Replace:**
   - In the dashboard workspace on the right, click on the sheet (visualization) you want to

replace. Tableau will highlight the selected sheet with a border.

2. **Ensure Context Awareness:**
   - Before replacing, consider the role of the current sheet in the dashboard. For example, is it linked to filters or actions? Make sure the replacement sheet fits the same purpose.

Finding the Replacement Sheet

1. **Locate the Replacement Sheet:**
   - In the Sheets list on the left panel, hover over the name of the sheet you want to use as a replacement.
2. **Use the Swap Sheets Button:**
   - When you hover over the replacement sheet in the Sheets list, a Swap Sheets button (usually represented by two arrows or a replace icon) will appear.

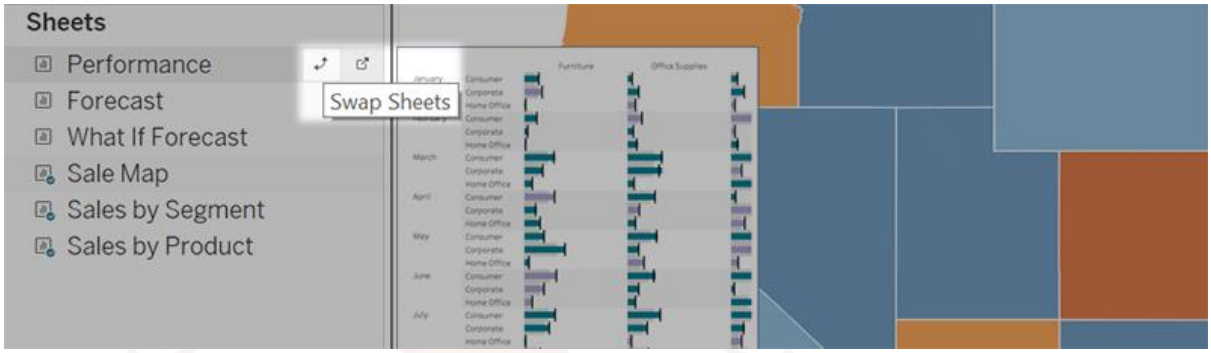
Swapping the Sheet

1. **Click the Swap Sheets Button:**
   - Click the button to instantly replace the selected sheet in the dashboard with the new one.
2. **Preserve Design Settings:**
   - Tableau retains most of the existing sheet's settings, such as padding, borders, background colors, and size.
   - If the content or dimensions of the replacement sheet differ significantly, you may need to make minor adjustments to the layout.

Post-Swap Adjustments

1. **Check Interactivity Settings:**
   - If the replaced sheet was part of an interactive filter or dashboard action, verify that these settings are still relevant and functional for the new sheet. Update them if necessary.
2. **Delete Unnecessary Items:**
   - If the replaced sheet included specific dashboard items (e.g., legends or filters) that are no longer relevant, remove these to avoid confusion or clutter.
3. **Adjust Layout and Appearance:**
   - Ensure the new sheet fits seamlessly into the dashboard layout. Resize or reposition as

needed to maintain a professional appearance.



Note: When you replace a sheet, Tableau retains any padding, border, or background color. However, if the content differs significantly, you may need to adjust the sheet size. You may also need to delete dashboard items specific to the previous sheet, such as filters, which become blank.

3.1. Adding Interactivity to Dashboards in Tableau

Interactivity in Tableau dashboards enhances user engagement and improves data exploration by allowing users to interact with visualizations and uncover deeper insights. Tableau provides several techniques to add interactivity, including using sheets as filters and adding dashboard actions. Here's a detailed explanation of these methods:

1. **Using Sheets as Filters**

   The Use as Filter option in Tableau transforms a visualization into an interactive control that filters other sheets within the dashboard.

   Steps to Enable Use as Filter:

   1. **Open the Dashboard:** Navigate to your dashboard in Tableau.
   2. **Select the Sheet:** Click on the sheet (e.g., bar chart, map, or scatter plot) that you want to use as a filter.
   3. **Enable the Option:**
      - In the upper-right corner of the sheet, click on the Use as Filter icon.
      - Once enabled, users can click on elements within the sheet (e.g., a specific bar or data point), and the corresponding data in other visualizations will be filtered automatically.

   Example Use Case:

   In a sales dashboard, clicking on a region in a map can filter related sales metrics in a bar chart

and customer demographics in a pie chart.

2. Adding Dashboard Actions

Dashboard actions allow users to create more advanced interactivity by linking multiple sheets, navigating between dashboards, or integrating external content like web pages.

Types of Dashboard Actions:

1. **Filter Actions:**
   - Allow selections in one visualization to filter data in other sheets.
   - Useful for creating drill-down capabilities (e.g., selecting a category to see sub-category details).
2. **Highlight Actions:**
   - Highlight related data points across different visualizations when users interact with a specific element.
   - For example, selecting a bar in a chart can highlight corresponding locations on a map.
3. **URL Actions:**
   - Enable users to click on a visualization and open a URL, such as a company website, product page, or external report.
   - Useful for dashboards that integrate external resources or link to detailed data.

Steps to Add Dashboard Actions:

1. **Access Action Settings:**
   - In Tableau Desktop, go to Dashboard > Actions from the top menu.
2. **Create a New Action:**
   - Click Add Action and choose the desired type (e.g., Filter, Highlight, or URL).
3. **Define Source and Target Sheets:**
   - Specify the source sheet where the action will be triggered and the target sheets where the action will apply.
4. **Set Trigger Conditions:**
   - Choose how the action is triggered (e.g., hover, select, or menu).
5. **Test the Action:**
   - Interact with the source sheet to ensure the action behaves as intended.

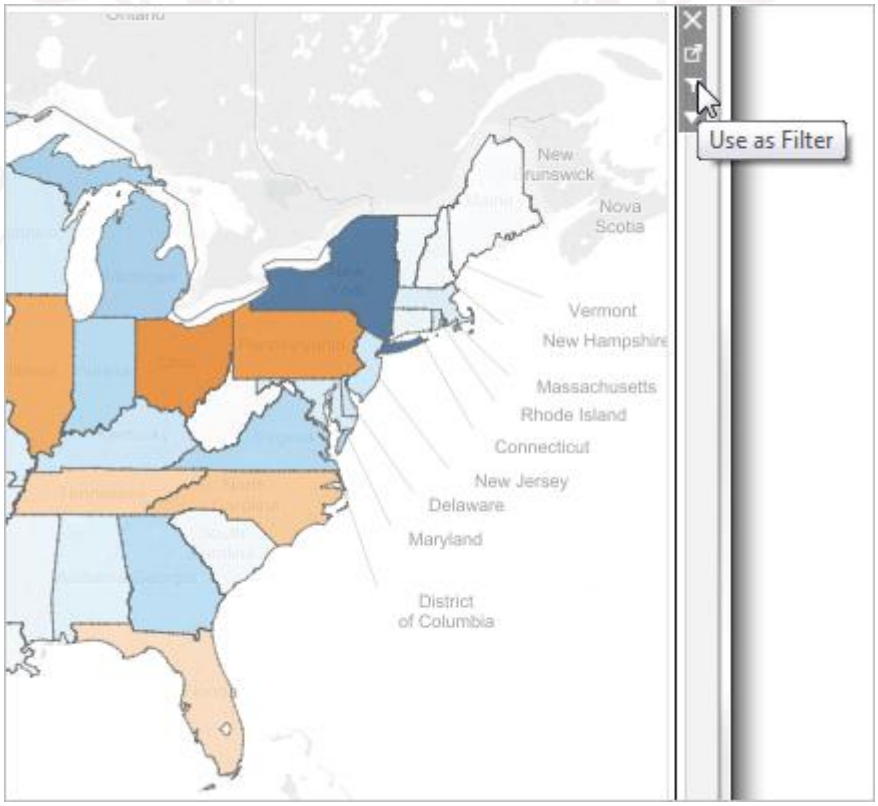
Example Use Case for Actions

- **Filter Action**: Clicking on a product category in a bar chart filters a line graph to show sales trends for that category over time.
- **Highlight Action**: Selecting a city on a map highlights its corresponding sales data in a table.
- **URL Action:** Clicking on a data point in a chart opens a webpage displaying detailed information about the selected item.

Benefits of Adding Interactivity

- **Improved User Engagement**: Users can explore data dynamically, making dashboards more engaging and insightful.
- **Custom Data Exploration**: Interactivity allows users to focus on data relevant to their needs, fostering deeper insights.
- **Streamlined Workflows**: Actions like navigation and filtering enhance efficiency by consolidating data exploration within a single dashboard.

By enabling the Use as Filter option and adding dashboard actions, Tableau allows users to create dynamic, interactive dashboards that empower decision-makers to uncover meaningful insights and make data-driven decisions effortlessly.



3.2. Adding Dashboard Objects and Setting Their Options in Tableau

In addition to integrating sheets into your dashboards, Tableau allows you to add various dashboard objects that enhance visual appeal, improve usability, and boost interactivity. These objects provide structure and supplementary elements, making dashboards more effective for users. Here's a detailed guide to each type of object and its options:

1. **Horizontal and Vertical Objects**

   Purpose:

   - These layout containers help organize dashboard components systematically, allowing you to group related objects.
   - They control the arrangement and resizing behaviour of items within the dashboard.

   Options and Usage:

   - **Horizontal Objects**: Align items side by side, such as placing a bar chart next to a pie chart.
   - **Vertical Objects**: Stack items on each other, such as placing a chart above a text box.
   - **Resizing**: Define how objects resize when the dashboard is viewed on different devices or when filters adjust content dynamically.

     Example: Group a chart, filter, and legend together using a vertical container to ensure consistent spacing and alignment.

2. **Text Objects**

   Purpose:

   - Text objects are used to provide context, instructions, headers, or additional explanations within your dashboard.

   Options and Usage:

   - **Formatting:** Customize font, size, alignment, and color to match your dashboard's design.
   - **Dynamic Text**: Use Tableau's capabilities to add dynamic text tied to data, such as filter selections or KPIs.

     Example: Add a header like "Quarterly Sales Performance" above your visualizations or include explanatory notes for filter usage.

3. **Image Objects**

   Purpose:

- Add logos, icons, or decorative elements to your dashboard to improve its visual appeal and branding.

Options and Usage:

- **Linking**: Attach specific target URLs to images so users can navigate to web pages when the image is clicked.
- **Alt-Text:** Add descriptive text for accessibility purposes or when the image fails to load.
- **Fitting Options**: Adjust how the image fits within its allocated space (e.g., fit, center, stretch).

  Example: Add your company's logo to the dashboard header and link it to the corporate website.

4. **Web Page Objects**

Purpose:

- Display an embedded web page directly within your dashboard.

Options and Usage:

- **Embedding Pages:** Add a live web page to provide supplementary content, such as news articles, product information, or external reports.
- **Security Settings:** Be mindful of web security settings, as some web pages do not allow embedding.

  Example: Display a webpage with livestock prices alongside a financial dashboard to give real-time context.

Note: Some organizations may restrict the use of Web Page objects for security reasons. If these options are unavailable, consult your Tableau administrator.

5. **Blank Objects**

Purpose:

- Blank objects allow you to fine-tune the layout by adding space or separating dashboard elements.

Options and Usage:

- **Spacing Adjustments**: Insert blank objects to create breathing room between tightly packed components.
- **Alignment Assistance**: Use blank objects to align items in a grid-like manner.

Example: Add blank spaces between filters and visualizations to make the dashboard look uncluttered.

6. **Navigation Objects**

Purpose:

- Navigation objects help users move seamlessly between dashboards, sheets, or stories within the workbook.

Options and Usage:

- **Button Design**: Add text or an image as the button for navigation.
- **Custom Appearance**: Specify border and background colors to align with the dashboard theme.
- **Tooltips**: Provide tooltips with additional information about the button's destination to enhance user experience.

  Example: Add a "Return to Overview" button to navigate from a detailed sales dashboard to the summary dashboard.

Best Practices for Using Dashboard Objects

1. **Maintain Consistency:** Ensure text formatting, spacing, and alignment are consistent across all dashboard elements.
2. **Optimize for Usability:** Use navigation buttons and text objects to guide users through the dashboard intuitively.
3. **Test Responsiveness:** Check how objects behave on different screen sizes using Tableau's device preview mode.
4. **Limit Web and Image Use:** Avoid overloading dashboards with web pages or images, as they can slow down performance.

By effectively using dashboard objects and customizing their options, you can create polished, professional, visually appealing, user-friendly, and interactive dashboards. These features help enhance the overall experience, making your Tableau dashboards more engaging and impactful.

- Download Objects enable users to quickly export an entire dashboard as a PDF, PowerPoint slide, or PNG image or to generate a crosstab from selected sheets. These objects share formatting options similar to those of Navigation objects.

Note: Crosstab downloads are only available after the dashboard is published on Tableau Cloud or Tableau Server.

- Extension Objects allow you to enhance your dashboard with custom functionalities or integrate it with external applications for extended capabilities beyond Tableau's native features.
- Pulse Metric Objects lets you embed pre-existing metric cards into your dashboard. These metrics are directly linked to the same published data sources used in the workbook, ensuring consistency and accuracy in the displayed data.

3.2. Adding and Copying Objects in Tableau Dashboards

Adding Objects

1. From the Objects section on the left panel of Tableau, drag the desired item to the dashboard workspace on the right.
2. Arrange and customize the object to fit your dashboard design.

Copying Objects

You can copy and paste dashboard objects to streamline the design process within the current dashboard or between dashboards in different workbooks or files. Copying can also be done across Tableau Desktop and Tableau in your web browser.

Limitations on Copying

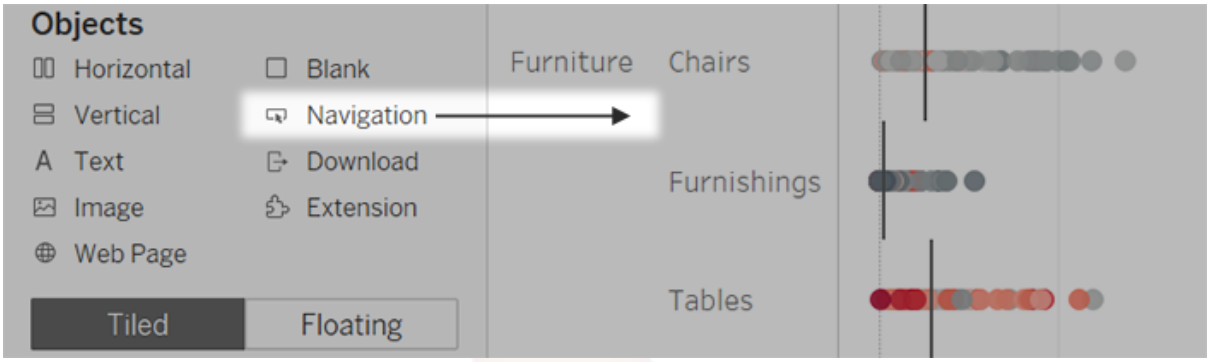
Certain items cannot be copied due to their dependencies or unique configurations:

- **Sheets in a Dashboard**: You cannot directly copy an entire sheet embedded in a dashboard.
- **Dependent Items**: Filters, parameters, and legends that rely on specific sheets cannot be copied.
- **Layout Containers with Restricted Items**: Containers holding non-copyable items, such as sheets or filters, cannot be copied.
- **Device Layout Objects**: Items in a device-specific layout are not eligible for copying.
- **Dashboard Titles:** Cannot be copied between dashboards.

Tips for Efficient Copying

- Use standard keyboard shortcuts (e.g., Ctrl+C/Ctrl+V or Command+C/Command+V) for quick copying and pasting.
- Ensure the target dashboard has compatible data sources or configurations to avoid errors when pasting copied objects.

By understanding these steps and limitations, you can efficiently duplicate dashboard components, saving time while maintaining consistency in design and functionality.



Copying Objects in Tableau

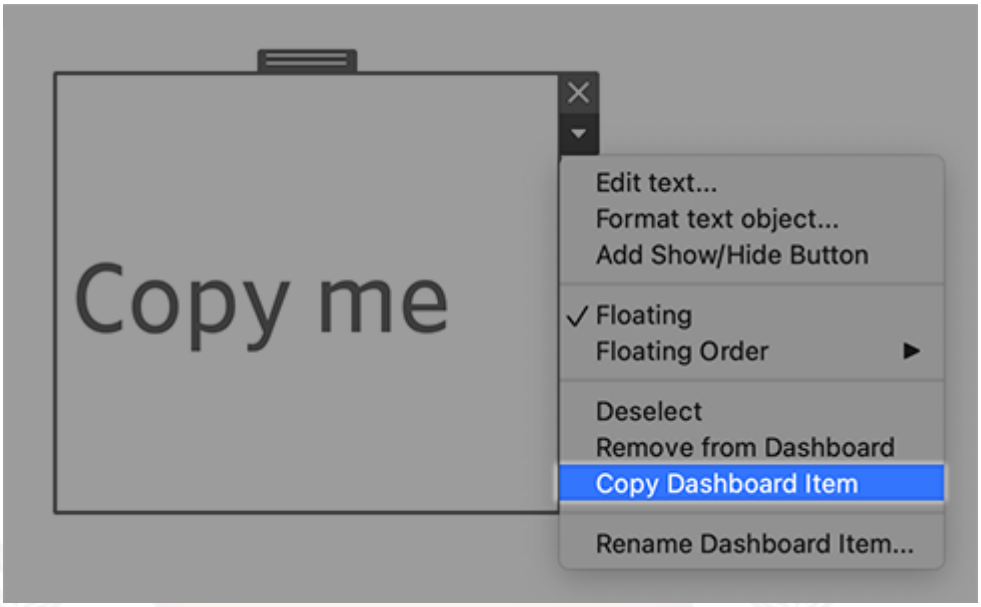
You can copy and paste dashboard objects either within the same dashboard or between dashboards in other sheets or files. Tableau also allows copying objects between Tableau Desktop and Tableau in a web browser.

However, certain items cannot be copied, including:

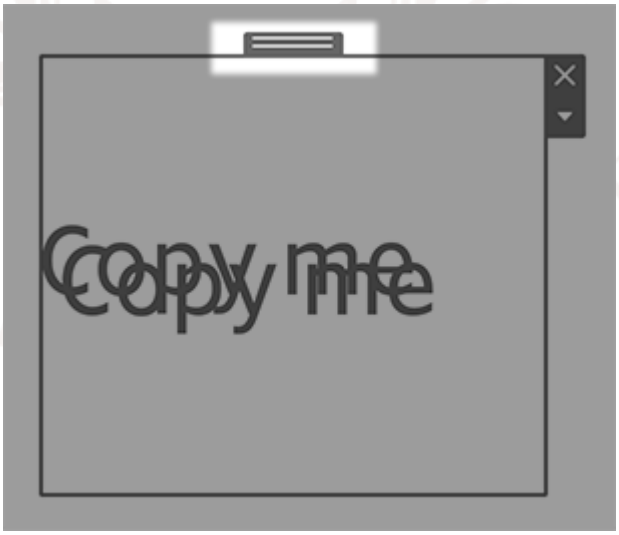
- Sheets embedded in a dashboard.
- Objects dependent on specific sheets, such as filters, parameters, and legends.
- Layout containers that include non-copyable items, like sheets or filters.
- Objects within a device-specific layout.
- Dashboard titles.

Tip: Besides using the menu commands for copying and pasting, you can utilize standard keyboard shortcuts specific to your operating system for faster operations.

1. Select a dashboard object; from the object menu, select Copy Dashboard Item. Or, from the main menu, select Dashboard > Copy Selected Dashboard Item.



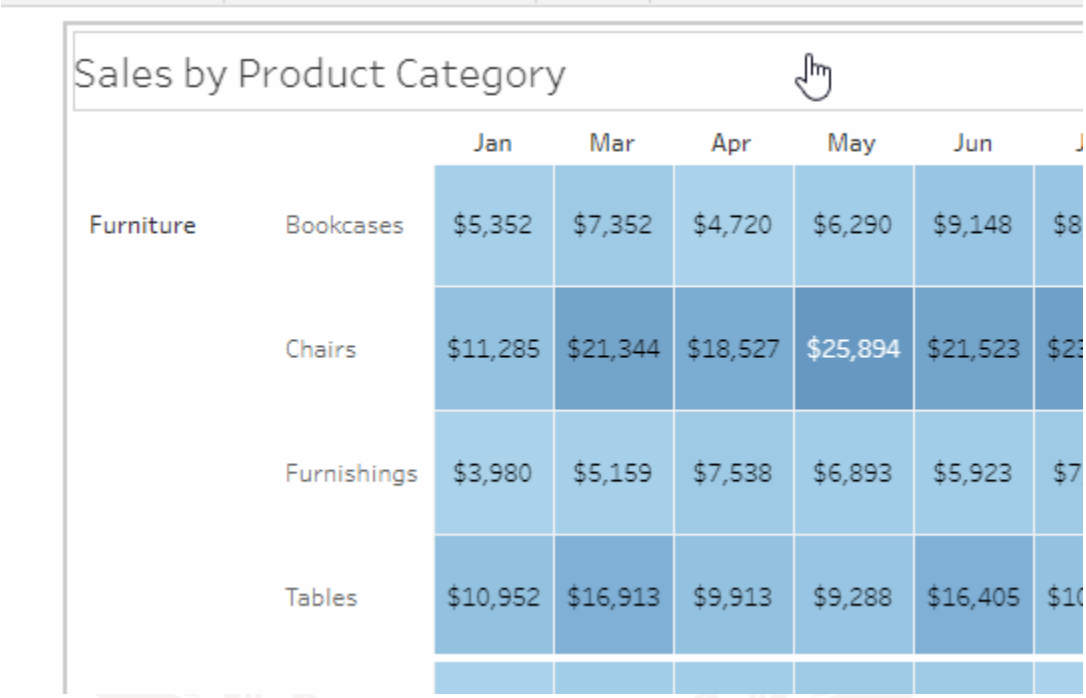
2. Navigate to the dashboard where you want to paste the object. To paste the object in the upper-left corner, ensure nothing is selected, or select an existing item to paste the object next to it.
3. In Tableau Desktop, go to File > Paste, or in a web browser, select Edit > Paste or use the standard keyboard shortcut for pasting.
4. The pasted object will appear 10 pixels below and to the right of the upper-left corner of the dashboard or the selected item. You can reposition it by dragging the handle at the top of the object.



Note: Copied Show/Hide buttons may target the original object or be disabled.

3.3. Setting Options for Dashboard Objects

To configure options for a dashboard object, click on the object to select it. Then, click the arrow in the upper corner to open the shortcut menu. The available menu options depend on the type of object selected.



a. Detailed Options for Image Objects

The Image object allows you to add images to your dashboard either by embedding them directly or linking to web-hosted images. Additionally, you can assign a URL to the image, enabling interactivity by redirecting users to the URL when the image is clicked.

Note: For security purposes, URLs for web-hosted images must include the HTTPS prefix. If the URL uses a different prefix, such as HTTP, use the Web Page object instead.

Steps to Add or Edit an Image Object

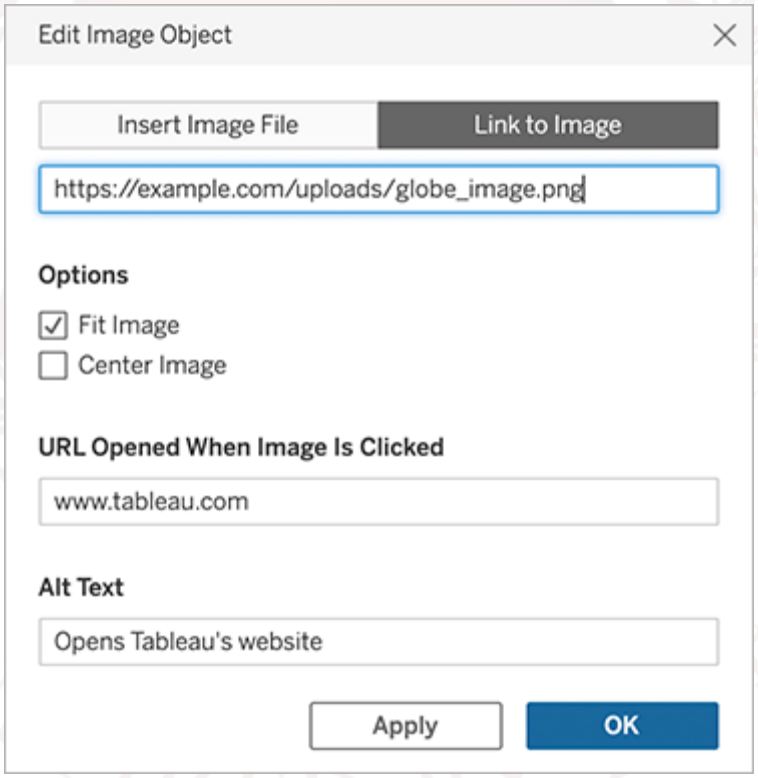
1. Drag an Image object from the Objects section on the left panel to your dashboard on the right. Alternatively, for an existing image, click the arrow in the upper corner and select Edit Image from the pop-up menu.
2. Choose one of the following options:
   - Insert Image File: Embed an image file directly into the workbook.
   - Link to Image: Add a link to a web-hosted image.

b. When to Use a Linked Image

Consider linking to a web-hosted image instead of embedding it when:

- The image file is very large, and your audience will access the dashboard in a browser. Linking prevents performance issues, as web-hosted images load dynamically without downloading each time the sheet opens.
- The image is an animated GIF. Embedded images in Tableau do not support animation, whereas linking allows animated GIFs to display properly.

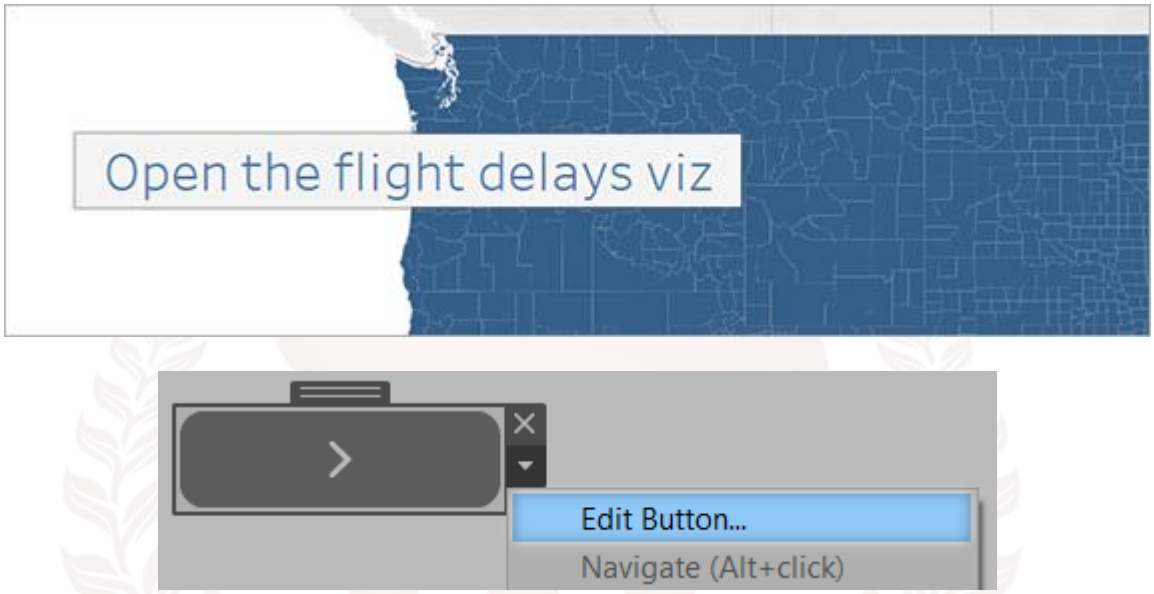
By customizing Image objects effectively, you can enhance the visual appeal of your dashboard while maintaining performance and interactivity.



1. If you are embedding an image, click Choose to locate and select the image file on your device. If linking to a web-hosted image, enter the image's URL.
2. Configure additional options, including image fitting, URL linking, and alt text. Alt text provides a description of the image for screen readers, enhancing accessibility for users with visual impairments.

c. Options for Navigation and Download Objects

Navigation and Download objects offer specialized settings that allow you to visually represent the navigation target or indicate the format of the downloadable file. These options help users intuitively understand the object's purpose and destination.



1. Click the object menu in the upper corner of the button and select Edit Button to modify its settings.
2. Choose an action for the button:
   - From the Navigate to menu, select a sheet outside the current dashboard.
   - From the Export to menu, select a file format for download.
3. Customize the Button Style by choosing either an image or text. Specify the image or text to display on the button and adjust the formatting options as needed.
4. Add Tooltip text to provide additional context when viewers hover over the button. This optional text is especially useful for image buttons. For instance, you might use a tooltip like "Open Sales Visualization" to clarify the destination of a navigation button represented by a miniature chart.

Note: In a published dashboard, clicking a button will perform the designated action (navigation or export). However, when designing the dashboard, you need to Alt-click (Windows) or Option-click (macOS) to activate the button.

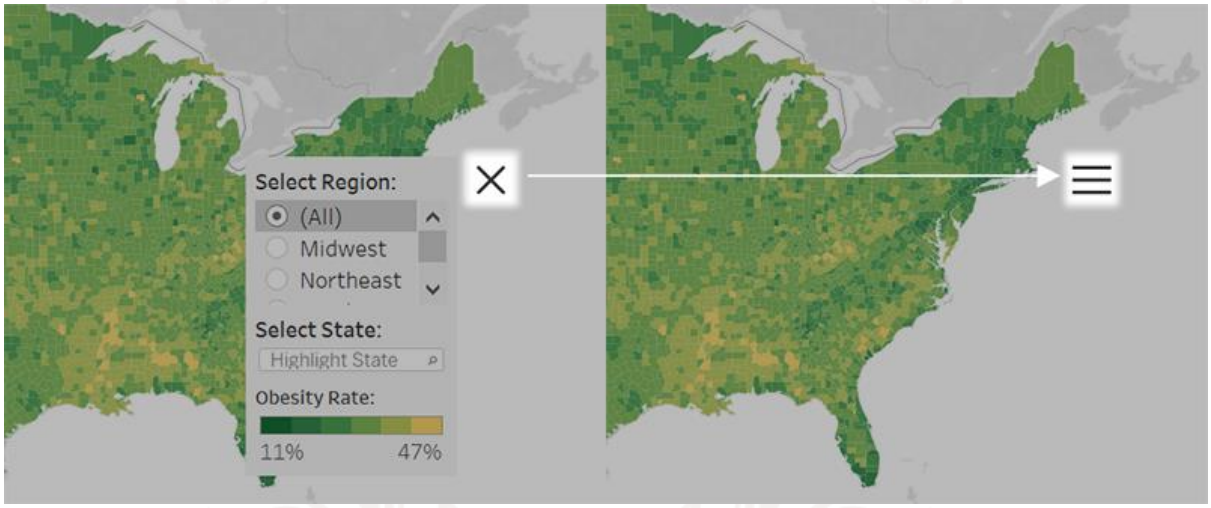
d. Show and Hide Objects Using Buttons

Show/Hide buttons allow dashboard viewers to toggle the visibility of specific objects, displaying them only when necessary.

How Hidden Objects Impact Layouts

- If the hidden object is floating, it simply uncovers any items layered beneath it.
- Show/Hide buttons are particularly useful for temporarily hiding floating groups of filters to reveal more of the underlying visualization.

This feature ensures a cleaner dashboard layout and offers an interactive way for users to focus on the most relevant data elements.



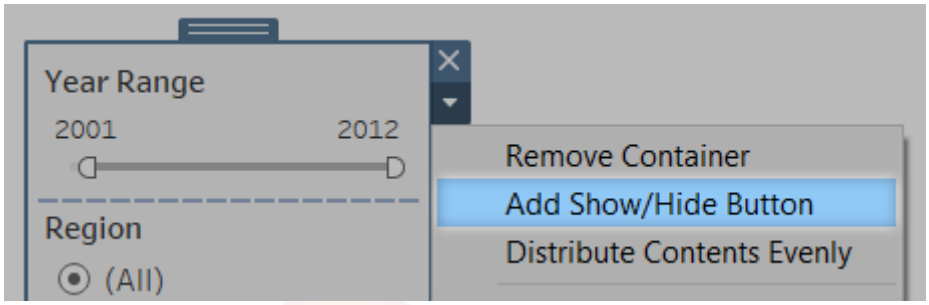
e. Behavior of Hidden Tiled Objects in Tableau

When a tiled object is hidden, the outcome depends on its position within the layout hierarchy:

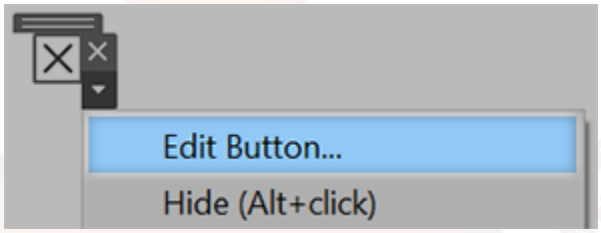
- **Horizontal or Vertical Layout Containers:** It is recommended to place objects you plan to hide within these containers. When an object is hidden, other objects in the container will automatically fill the vacant space, maintaining a seamless layout.
- **Top-Level Tiled Layout Container:** If the hidden object is in the top-level tiled container, it will leave blank space behind, as no other objects can fill the gap.

f. Adding and Configuring a Show/Hide Button

1. **Select an Object**: Choose the dashboard object you want to show or hide.
2. **Add the Button**: Click the pop-up menu in the upper corner of the selected object and choose Add Show/Hide Button to enable toggling of the object's visibility.



3. From the button menu, choose Edit Button.



4. **Set the Following Options:**
   - **Dashboard Item to Show/Hide:** Choose the object that the button will control. An object can only be targeted by one Show/Hide button at a time. If you plan to assign another button to the object, set this option to **None**.
   - **Button Style:** Select whether the button will display an image or text.
   - **Button Appearance:** Customize how the button looks in both states—when the target item is visible and when it is hidden. Use the **Item Shown** and **Item Hidden** options to specify different images or text for each state.
   - **Tooltip Text:** Add explanatory text that appears when users hover over the button. For example, for a container holding filter menus, you could enter, "Show or hide filters."
5. **Adjust the Button:** Drag the button to the desired position or resize it as needed to align with your dashboard layout.

Note: In a published dashboard, clicking the Show/Hide button will toggle the object's visibility. However, when designing the dashboard, you must use **Alt-click** (Windows) or **Option-click** (macOS) to perform the toggle action.

g. Enhancing Security for Web Page Objects

If your dashboard includes Web Page objects, consider the following security practices to protect users and ensure functionality:

- **Use HTTPS Protocol Whenever Possible:**

- Include URLs with HTTPS (e.g., https://) to encrypt the connection between your dashboard and the web page.
- If Tableau Server is configured to use HTTPS and an HTTP URL is used, browsers may block the web page from displaying.
- If no protocol is specified, Tableau assumes HTTP, which may result in less secure connections.

Security Options for Web Page Objects (Tableau Desktop Only):

Tableau Desktop provides additional security settings for Web Page objects, allowing you to fine-tune access and behavior to align with organizational security policies.

h. Configuring Web View Security Settings in Tableau

To set security options for dashboard web views, go to Help > Settings and Performance > Set Dashboard Web View Security and configure the following options. Note that in some organizations, these settings may be managed globally by a Tableau administrator.

Important: Changes made to these settings apply to both existing and newly created Web Page objects.

- **Enable JavaScript**: Allows the use of JavaScript in web views. Disabling this option may cause web pages requiring JavaScript to malfunction within the dashboard.
- **Enable Plug-ins**: Enables any plug-ins the web page relies on, such as Adobe Flash or QuickTime.
- **Block Pop-ups**: Blocks pop-up windows when selected.
- **Enable URL Hover Actions**: Allows URL hover actions to display. For more details, refer to Tableau's documentation on URL Actions.
- **Enable Web Page Objects and Web Images**: Permits Web Page and Image objects to display linked target URLs. If this option is disabled, these objects will remain in the dashboard but will not display any content.

4. COMBINING MULTIPLE CHARTS INTO INTERACTIVE DASHBOARDS IN TABLEAU

Tableau enables users to combine multiple charts into a single interactive dashboard, creating an integrated workspace for data exploration and storytelling. This approach facilitates comprehensive analysis by allowing users to interact with multiple datasets seamlessly.

Step-by-Step Guide:

1. **Create Individual Sheets:**
   Begin by designing individual visualizations, such as bar charts, line graphs, or maps, on separate sheets in Tableau. Each sheet should represent a distinct data perspective.
2. **Open the Dashboard Workspace:**
   Click the **New Dashboard** button located at the bottom of the Tableau interface. This action opens a blank dashboard workspace where visualizations and additional elements can be integrated.
3. **Drag and Drop Visualizations:**
   From the **Sheets** panel on the left, drag your selected charts into the dashboard workspace. Use Tableau's grid-based layout tools to organize and position them according to your design preferences.

Adding Interactive Elements

- **Filters:** Include filter widgets in your dashboard to allow users to refine data across multiple charts simultaneously, enabling a more focused analysis.
- **Actions:** Configure dashboard actions like filter actions, which let users click on a chart element to filter related data in other visualizations, or highlight actions, which emphasize related data points across charts.
- **Parameters:** Use parameters to give users dynamic control over certain aspects of the dashboard, such as selecting metrics or adjusting date ranges.

Customizing Layout and Appearance

Leverage Tableau's layout tools to fine-tune the dashboard's padding, borders, and background colors for a polished and professional design. Ensure the layout is optimized for the target screen size and audience to provide the best user experience.

Testing Interactivity

Interact with the dashboard to verify that all filters, actions, and other interactive elements are working as intended, ensuring a smooth user experience.

Benefits of Combining Charts into Dashboards

- **Enhanced Insights:** Offers a comprehensive view of data from different perspectives, allowing for deeper analysis of complex relationships.
- **User Engagement:** Encourages active exploration of data, making the dashboard more interactive and engaging compared to static reports.
- **Efficient Decision-Making:** Centralizes all relevant data in one place, enabling stakeholders to make faster and more informed decisions.

Example Use Case

In a sales dashboard, you might include:

- A map visualization to display regional sales distribution.
- A bar chart comparing performance across product categories.
- A line graph showing sales trends over time.

Interactive elements like filters and actions would let users drill down into specific regions or products, creating a seamless and insightful data exploration experience.

By integrating multiple charts into an interactive dashboard, Tableau converts raw data into actionable insights, empowering users to make data-driven decisions with ease.

5. ADDING INTERACTIVITY WITH FILTERS

5.1. Filter Actions in Tableau

Applies to: Tableau Cloud, Tableau Desktop, Tableau Server

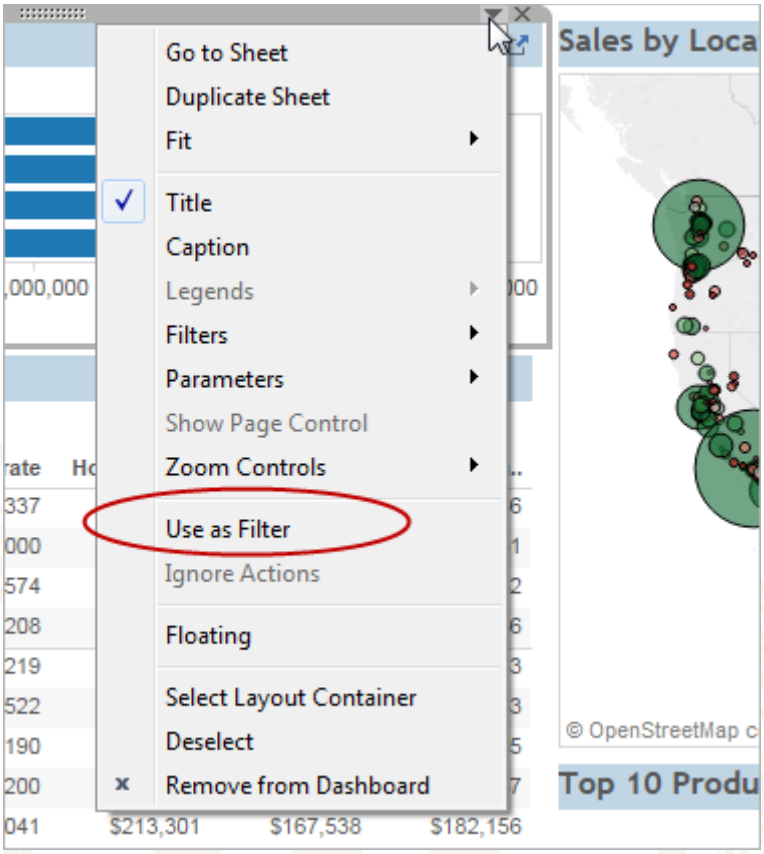
Filter actions enable communication between worksheets by sending information from a selected mark in one sheet to another sheet displaying related data. These actions work by transferring data values from the source fields to the target sheet as filters.

For instance, in a visualization showing house sales prices, selecting a specific house could trigger a filter action to display comparable houses in another view. In this example, the source fields might include data like sales price and square footage.

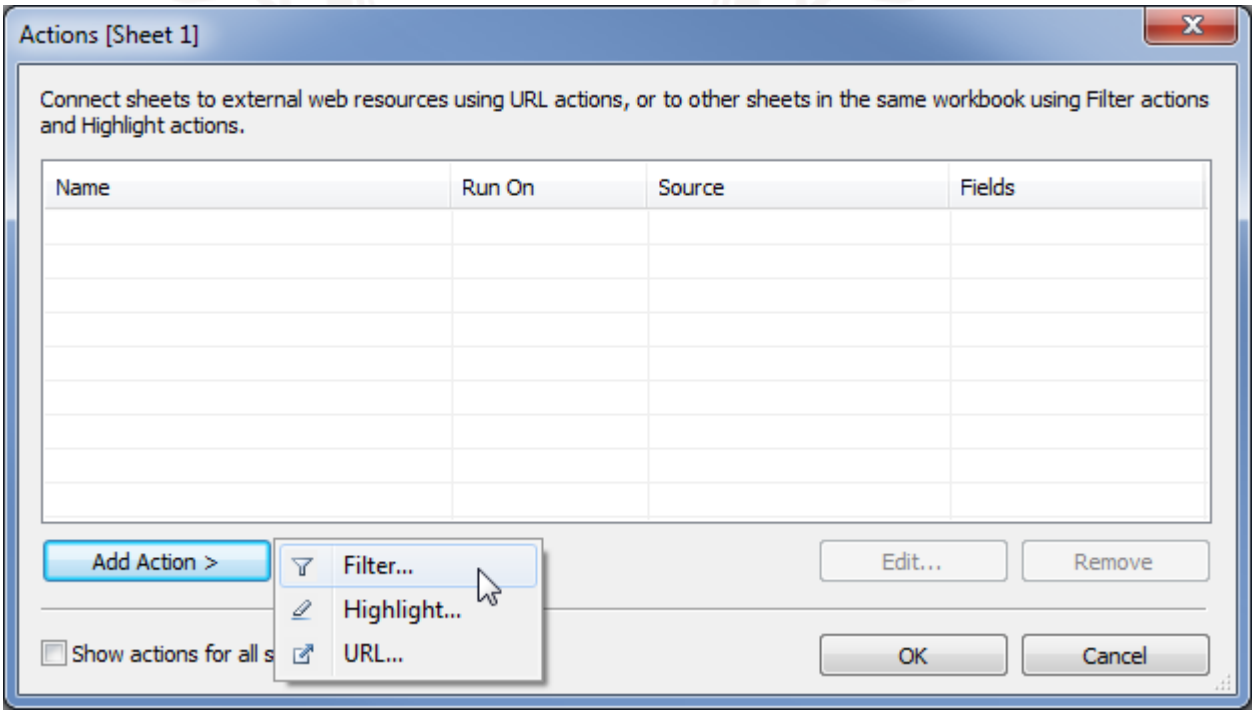
How to Create or Edit a Filter Action

1. Choose a Starting Point:
   - On a worksheet, go to Worksheet > Actions from the menu.
   - On a dashboard, navigate to Dashboard > Actions.
   - Alternatively, use the Use as Filter option from the drop-down menu of a dashboard sheet.
2. Manage Actions:
   - When using the Use as Filter option, a default filter action is generated, which will appear in the Actions dialog box with the term "generated" in its name.

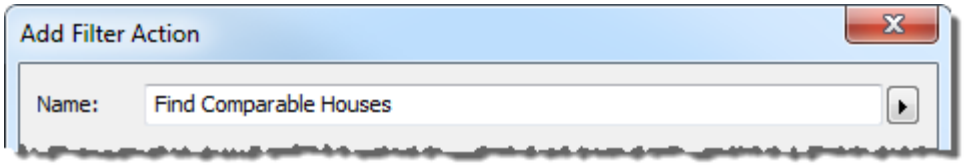
   By using filter actions, you can create dynamic, interconnected dashboards that provide a deeper and more interactive data exploration experience.



3. In the Actions dialog box, click Add Action, and then select Filter. Or select an existing action, and choose Edit.

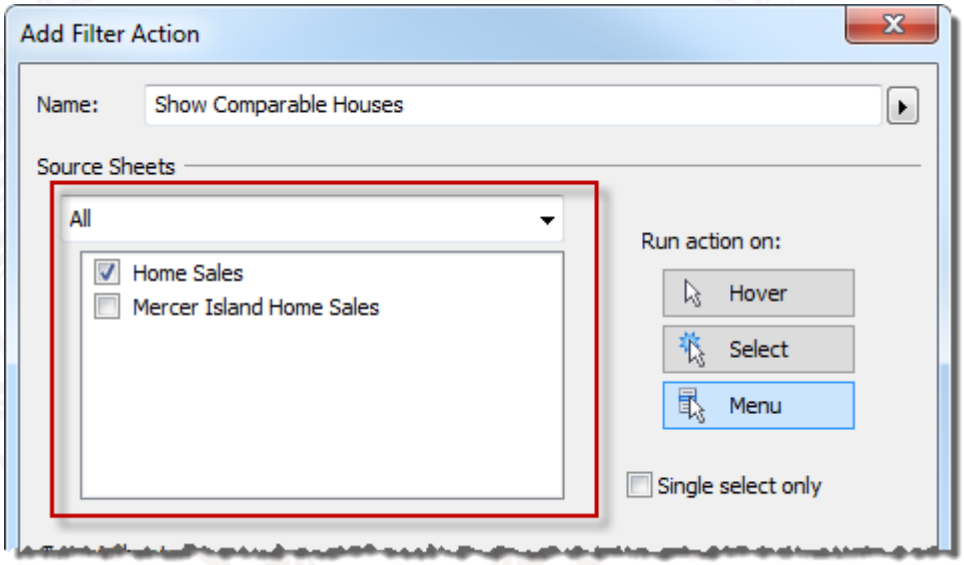


4. Specify a name for the action.

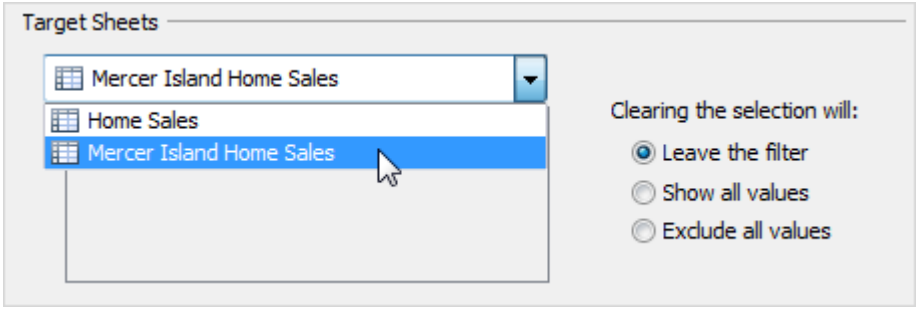


Tip: If you plan to launch an action from a tooltip menu, use a descriptive name so users will understand the action's purpose. From the menu to the right of the Name box, you can insert variables that are drawn from the values of selected fields.

5. Select a source sheet or data source. If you select a data source or dashboard, you can select related sheets you want to launch the action from.



6. Specify how the action runs:
   - Hover - Runs when you mouse over marks in the view.
   - Select - Runs when you click marks in the view. If you want to avoid running the action when multiple marks are selected, also select Single-select only.
   - Menu - Runs when you right-click (Windows) or Control-click (macOS) a mark in the view, and then click an option in the tooltip menu.
7. Select a target sheet. When you select a dashboard, you can select one or more sheets within it.



8. Specify what happens when the selection is cleared in the view:
   - **Leave the filter** - Continues to show filtered results on the target sheets. (In web-authoring mode, this is labeled Keep filtered values.)
   - **Show all values** - Changes the filter to include all values.
   - **Exclude all values** - Changes the filter to exclude all values. This option is useful when you're building dashboards that only show some sheets if a value in another sheet is selected.
9. Specify the data that you want to show on the target sheets. You can filter on All Fields or Selected Fields.
10. If you chose Selected Fields, click a drop-down menu in the Source Field column, and select a field. Then select a target data source and field.



Note: Filter actions that depend on a user function, such as USERNAME(), will not work because row-level security restricts access to the data.

Understanding Available Target Fields in Filter Actions

In the Add Filter dialog box, the fields shown in the Target Field drop-down list are restricted by the data type of the Source Field. For example, if the source field is a text field, only text fields will appear as options for the target field.

If you are working with a relational data source, you can create links between sheets across different data sources, even if the field names are not identical. For instance, a Latitude field in one data source can be associated with a Lat field in another by selecting the appropriate fields from the drop-down lists in the Add Filter dialog box.

When using a multidimensional data source, the source and target sheets must rely on the same data source, and the field names for both the source and target must match exactly. (Note: Tableau supports multidimensional data sources only on Windows.)

6. HIGHLIGHTS

Highlights in Tableau are a feature that helps users emphasize specific data points or relationships in visualizations, making it easier to identify patterns, trends, and outliers. Highlighting dynamically interacts with selected marks or fields, providing visual cues that focus on relevant data while dimming the rest of the view.

Ways to Use Highlights in Tableau

1. **Highlight Actions**:
   - **What They Do:** Allow users to select a data point in one visualization and see related data highlighted in another visualization.
   - **Example:** Selecting a region in a map could highlight sales data for that region in a bar chart.
   - **How to Set Up:**
     - Go to **Dashboard > Actions > Add Action > Highlight**.
     - Specify the source and target sheets and configure the fields to be highlighted.
2. **Highlight Buttons**:
   - **What They Do:** Add a button to toggle highlights on or off.
   - **Example:** Use a button to highlight products exceeding a certain sales threshold in a dashboard.
3. **Highlighting Legends**:
   - **What They Do:** Enable users to click on a legend item to highlight marks in the view that correspond to that category.
   - **Example:** Clicking on a "Technology" category in a legend dims non-technology data in a sales visualization.
4. **Manual Selection**:
   - **What It Does:** Users can manually hover over or click on marks in the visualization to highlight related data points.
   - **Example:** Hovering over a specific data point in a scatter plot highlights all associated marks in other views.

6.1. Benefits of Highlighting

- Enhanced Focus: Helps users zero in on specific data points or categories without losing

context.

- Improved Exploration: Facilitates deeper analysis by dynamically connecting related data points across multiple views.
- Interactive Storytelling: Engages users by making visualizations more interactive and intuitive.

6.2. Best Practices

- Use highlighting sparingly to avoid overwhelming users with too much visual emphasis.
- Ensure that highlights are meaningful and add value to the analysis.
- Combine highlighting with other interactive elements, such as filters and tooltips, for a comprehensive exploratory experience.

Method	Benefits	When to Use
Select Marks	Manually select a group of marks to highlight in a view. Your selection is saved with the workbook.	When you want to manually highlight a selection of marks and dim all others. Works well with small domains or views with limited data.
Legends	Supports one-way and two-way highlighting on color, size, or shape. Your selection is saved with the workbook and can be included in dashboards.	When you want to focus on specific members and dim all others. Works well with small domains or views with limited data.
Highlighter	Search or select data points using keywords or a drop-down list. Maintains context while highlighting. Updates automatically with data changes.	When you want to highlight marks for a discrete field and make ad hoc comparisons. Works well with large domains and large datasets.
Actions	Define criteria to highlight data. Specify source and target sheets, and fields for highlighting. Multiple actions can be combined (e.g., filter + highlight).	When building interactive exploration into a dashboard and highlighting data points using specific fields.

Highlights in Tableau are a powerful way to direct attention, uncover insights, and create engaging, interactive dashboards.

Legend highlighting in Tableau allows you to direct users' focus to specific marks in a view by highlighting those associated with a selected legend item, while dimming the rest.

You can enable either one-way or two-way highlighting:

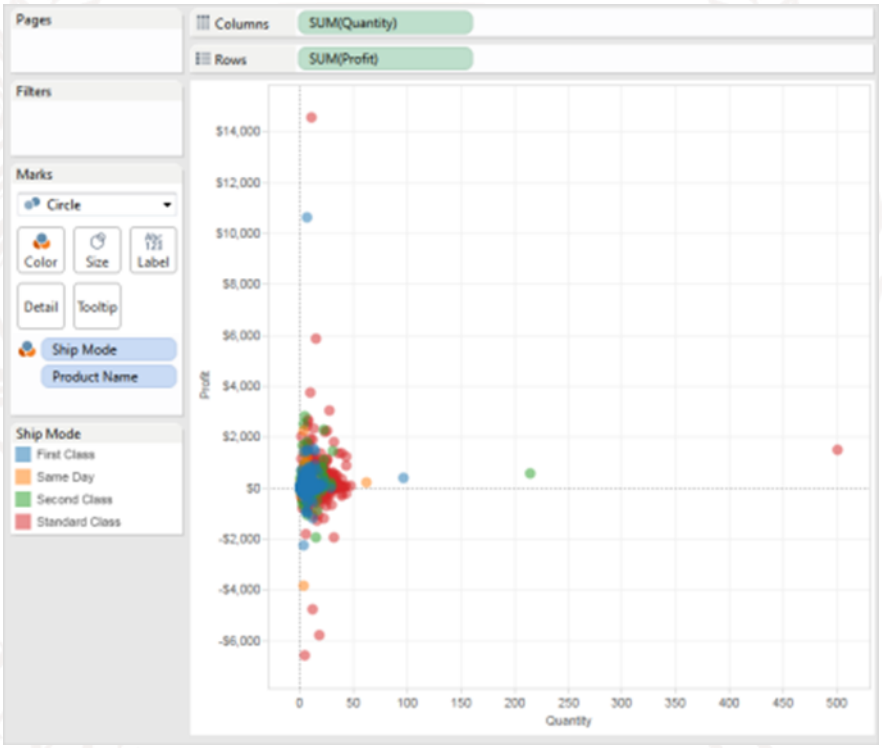
- **One-way highlighting** allows you to highlight marks based on the values in the legend.

- **Two-way highlighting** is the default mode, where selecting marks in the view will also highlight the corresponding legend item, and vice versa.

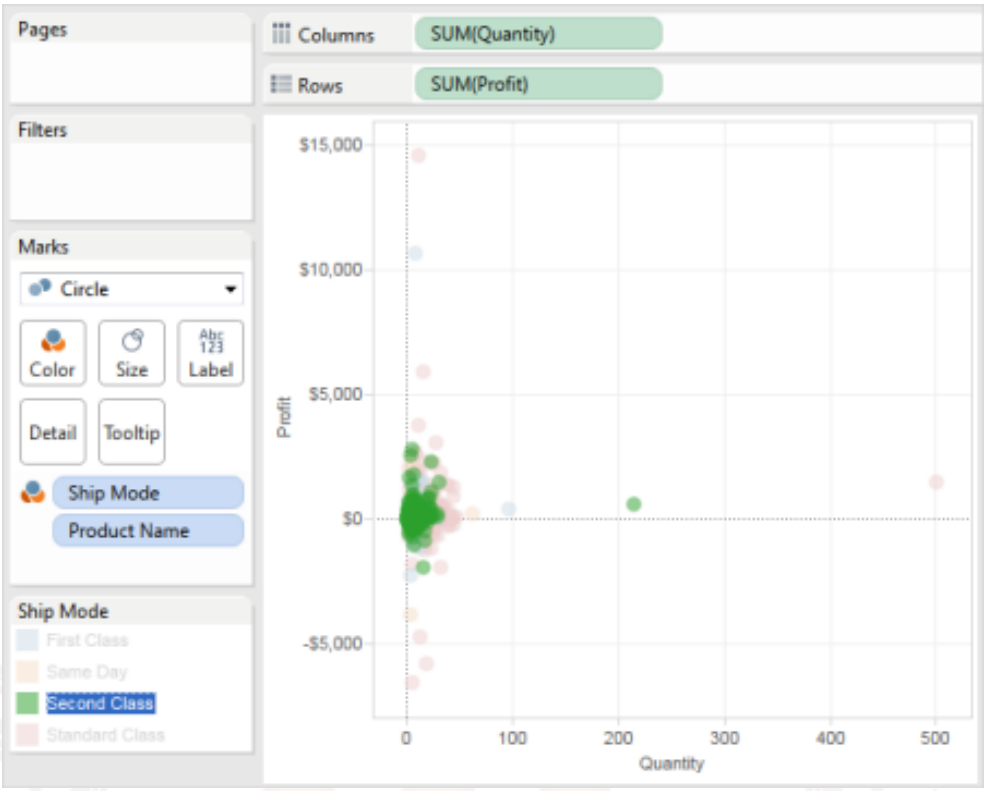
Highlighting can be turned on or off for the entire workbook or a specific sheet using the toolbar, which also hides the highlight icon in the legend.

For example, in the views below, the first one uses the standard color legend functionality where marks are coloured based on their shipping mode. The second view uses legend highlighting to emphasize the products shipped via Second Class.

Normal Color Legend



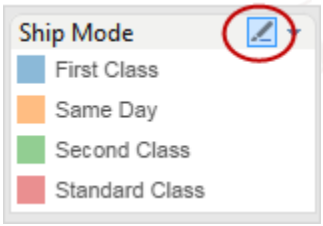
Color Legend Highlighting Enabled



To use color legend highlighting in Tableau, you can control the highlighting behavior through the legend card menu.

Turning on Legend Highlighting:

1. Click the Highlight button at the top of the legend.
   - In Tableau Desktop, you can also turn it on by selecting Highlight Selected Items from the legend card menu.
   - By default, color legend highlighting is enabled.



2. Select an item in the color legend.
   - Once enabled, you can focus on specific data by selecting different items in the color legend. A highlight action is created, which you can adjust in the Actions dialog box.

Turning off Legend Highlighting:

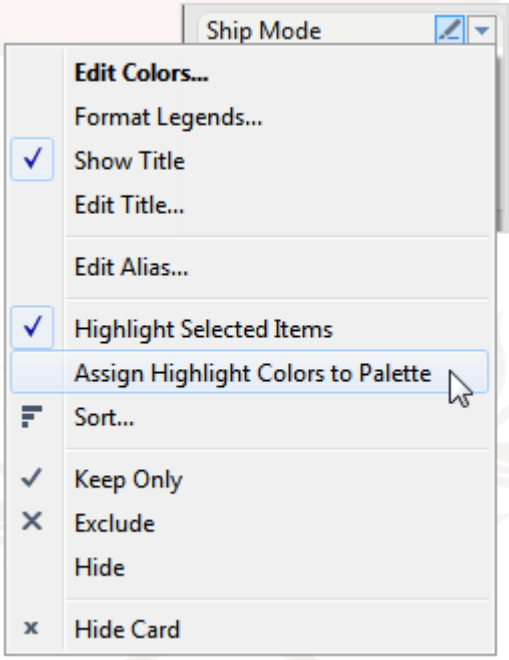
- Click the Highlight button again to change the mode to one-way, where you can use the legend

to highlight matching marks in the view.

- In Tableau Desktop, you can also turn it off by choosing Highlight Selected Items from the legend card menu.
- Disabling legend highlighting removes the action from the Actions dialog box.

Assigning Highlight Colors to the Palette:

- If you like how the view is highlighted and want to preserve a specific highlight, you can assign the highlight colors to the color palette.
  - To do this, select Assign Highlight Colors to Palette in the color legend card menu. This will replace the original color legend with the highlight colors as the new color palette.



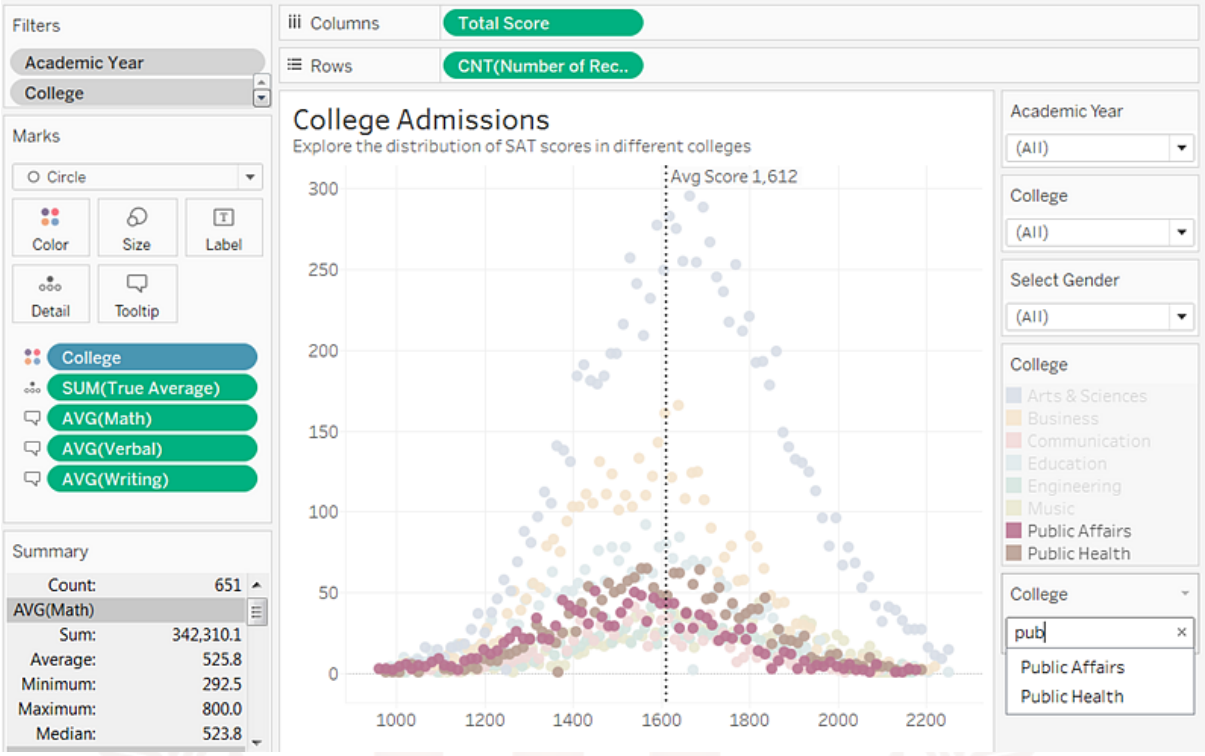
Highlight Data Points in Context

When working with a large dataset in Tableau, you may want to highlight specific marks or groups of marks while keeping the context of the rest of the data visible.

To do this, you can enable the Highlighter for one or more discrete fields in your view that impact the level of detail. The Highlighter allows you to search for matching data points using keywords. As you type, Tableau instantly highlights the marks that match or partially match your search term. If the underlying data is updated, the highlighted results will automatically reflect the changes.

For example, if the Highlighter is enabled for the College field and you enter the partial keyword "Public," Tableau highlights the matching marks, such as Public Affairs and Public Health, within the

view.



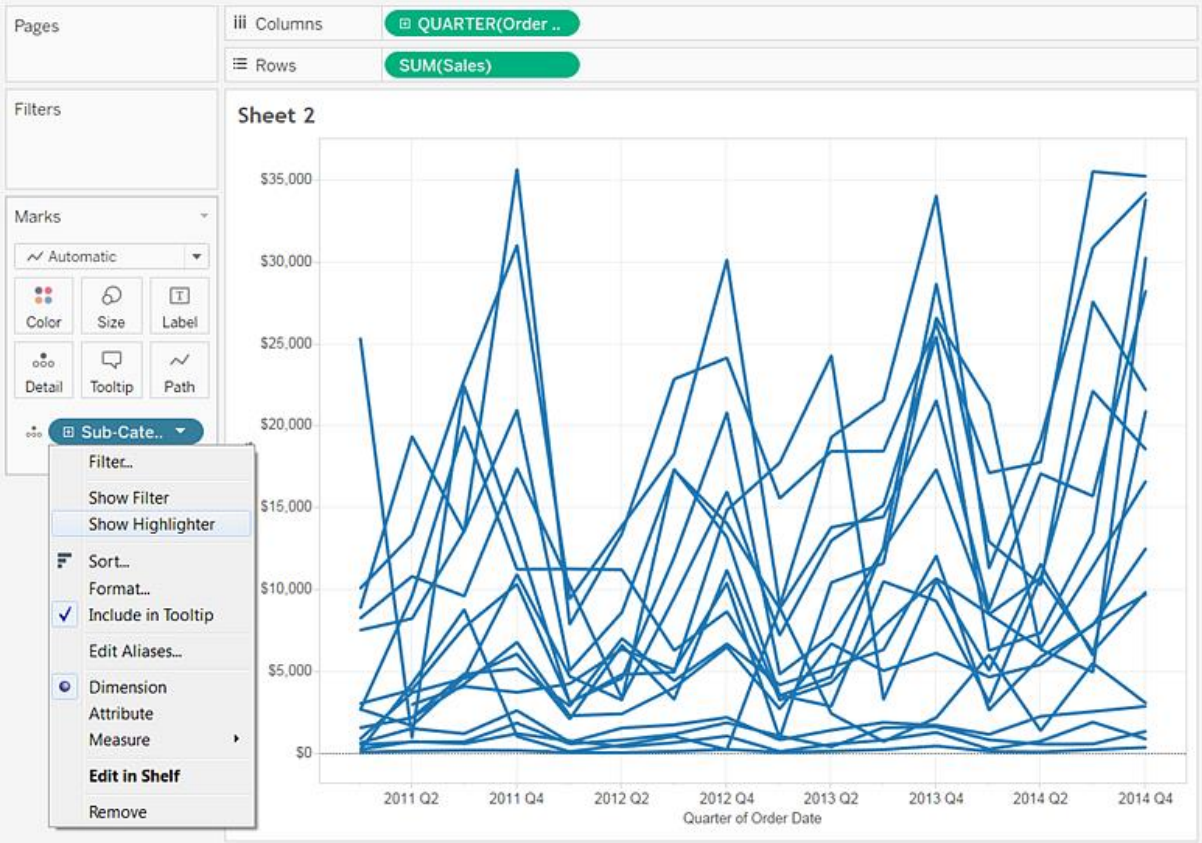
You can enable highlighters for multiple discrete fields in your view, but only one value set can be highlighted at a time. To include the highlighter option in dashboards and stories, activate it in the worksheet before adding it to the dashboard. Alternatively, after adding the worksheet to the dashboard, select it and go to Analysis > Highlighters from the top menu to choose from the available fields.

Tableau Desktop only: To change the format of the highlighter card, go to Format > Highlighter after it is turned on.

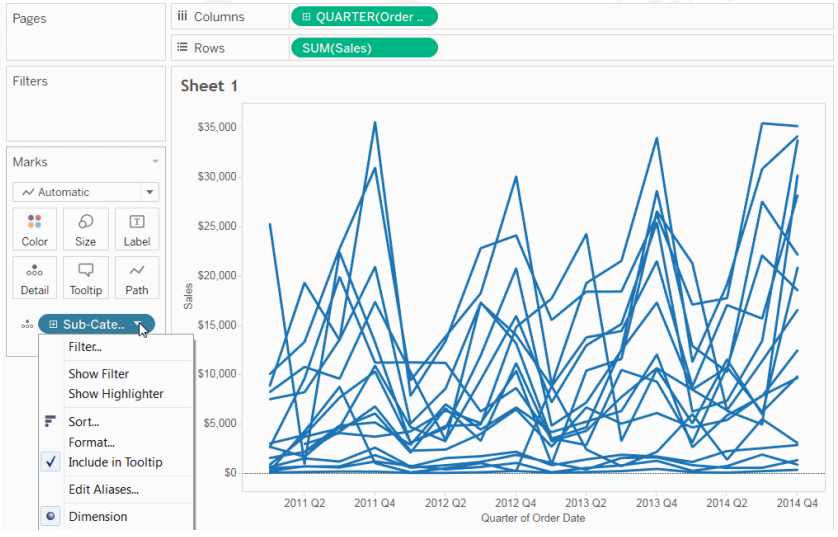
Highlighters can work across multiple worksheets in a dashboard, as long as the same field is included in each. If the field is missing in any worksheet, no match will be found, and those values will be dimmed. If multiple highlighters are present, only the most recently used highlighter will be active.

To turn on highlighting:

1. Right-click on a discrete field included in the view that impacts the level of detail, then select Show Highlighter from the context menu.



2. In the highlighter dialog box, do one of the following:
   - Type a keyword to search for matching marks to highlight.
   - Type a partial keyword to find all relevant matches containing the search text.
   - Choose an item from the drop-down list, selecting one item at a time.
   - Hover over items in the drop-down list to highlight marks in the view for an ad hoc analysis experience.



3. Repeat steps 1 and 2 to add additional highlighters.

   You can also show mark labels on highlighted marks. To turn on mark labels, on the Marks card, click Label, select the Show mark labels check box and select Highlighted in the Marks to Label section.

7. DASHBOARD ACTIONS

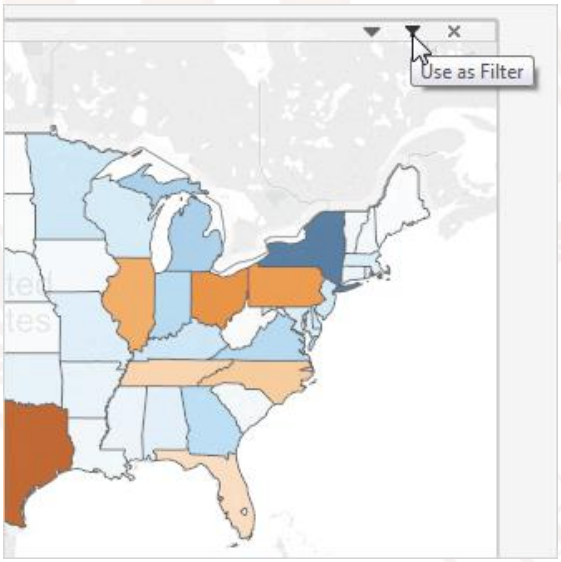
Dashboards often contain multiple views, and a single filter or highlight action can impact all of them. Dashboards can also include web page objects, which can be targeted with interactive URL actions.

Using a Single View to Filter Other Views in a Dashboard:

For example, you have a dashboard with three views about profitability: a map, a bar chart, and a table of customer names. You can set one view, such as the map, to act as the controlling filter. When users select a region on the map, the other views are filtered to display data relevant to that region.

Steps to use a view as a filter:

1. Select the view you want to use as a filter on the dashboard.
2. From the view's shortcut menu, select Use as Filter, or click the Use as Filter icon.

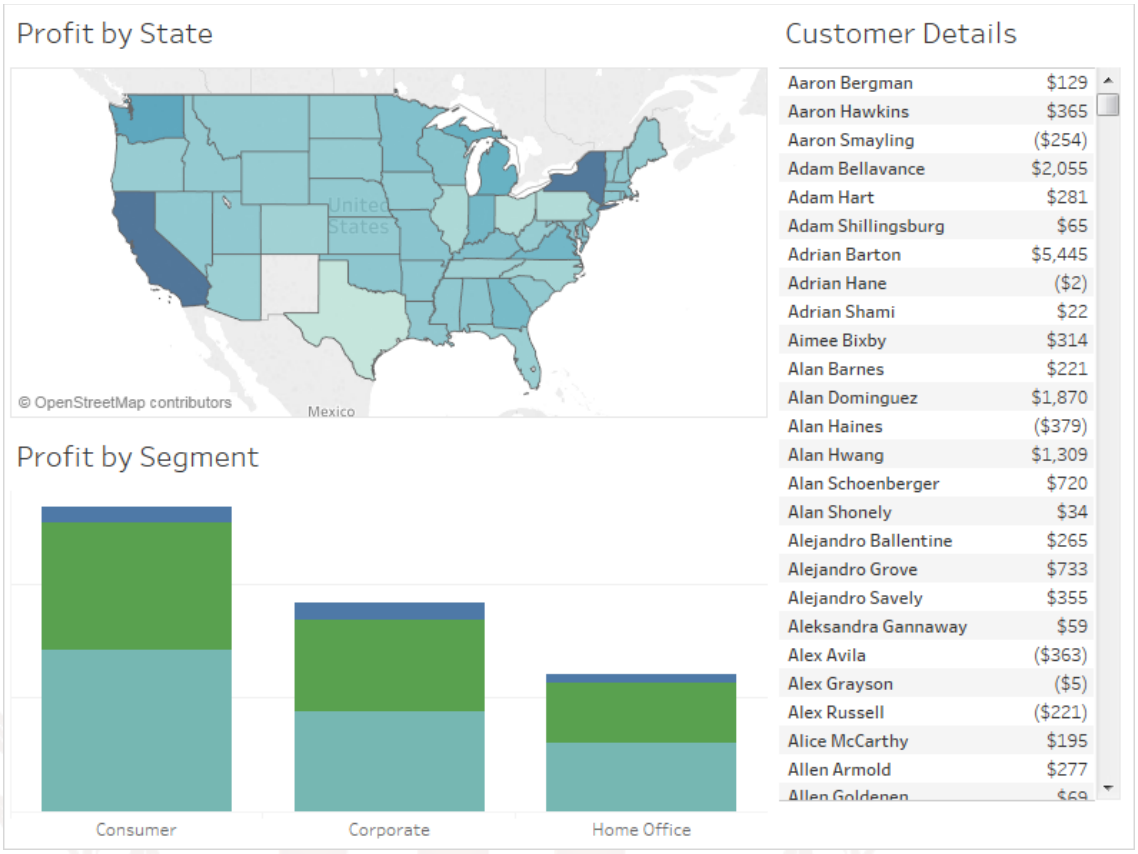


Using Multiple Views to Filter Other Views in a Dashboard:

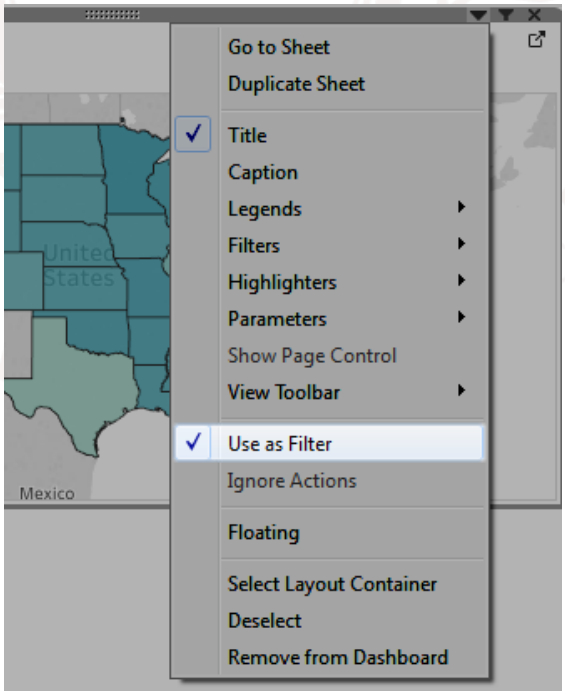
Just like using a single view to filter other views in a dashboard, you can also use multiple views as filters. The key is to set up these views as filters while disabling their ability to be filtered themselves.

Steps:

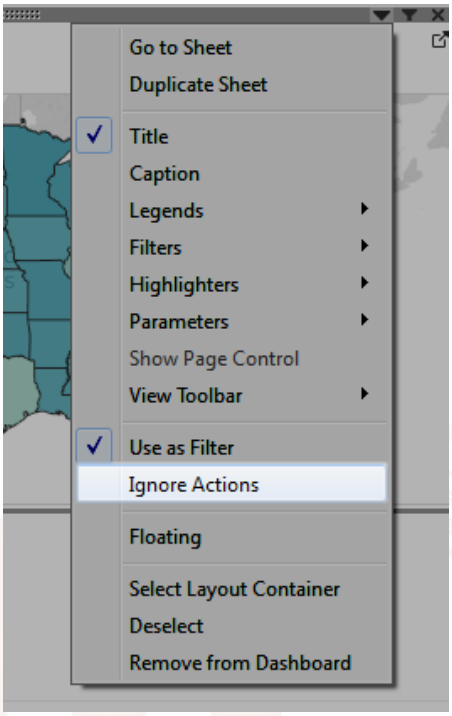
1. Create or open a dashboard with at least three views.



2. Select the first view that you want to use as a filter (such as a map), and from its shortcut menu, select Use as Filter.

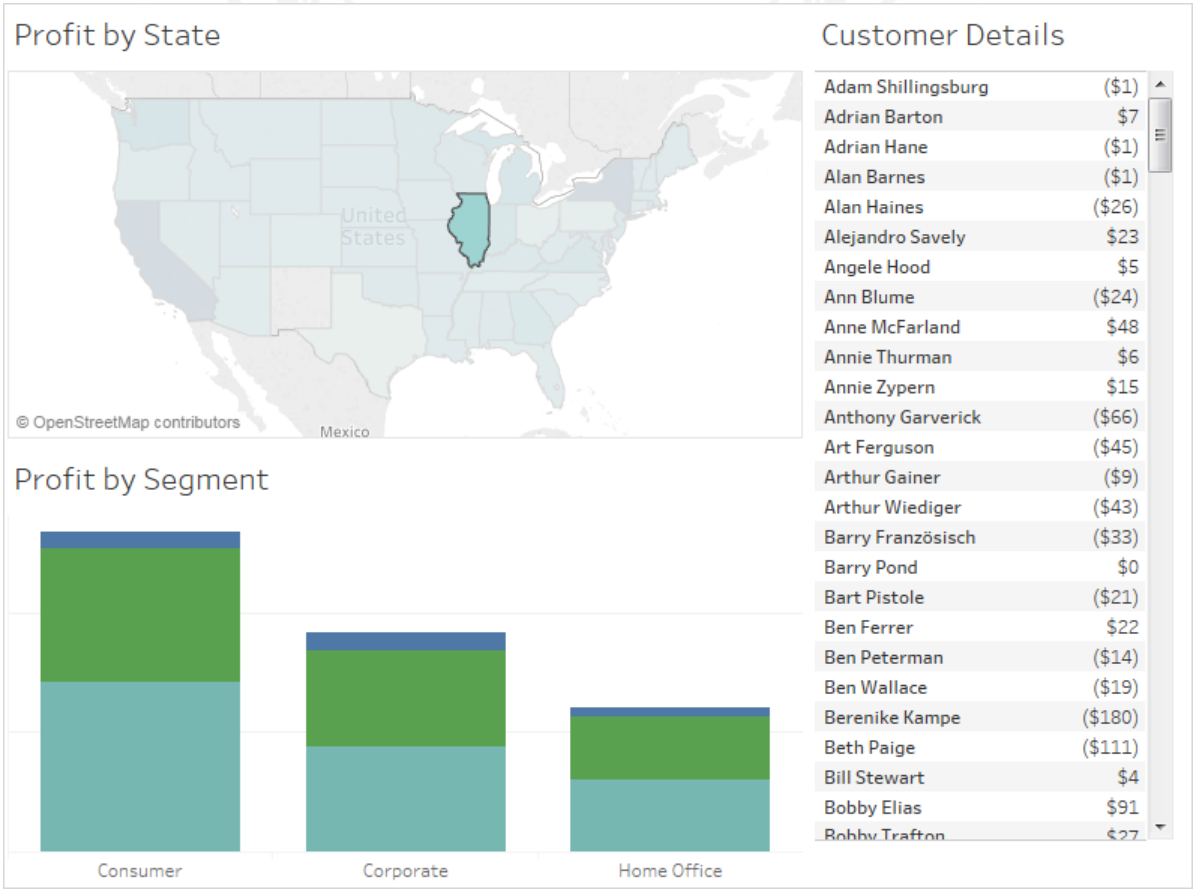


3. Open the same view's shortcut menu again and select Ignore Actions. This ensures that other filter actions, including the one you'll create next, will not affect this view.



4. Repeat the steps 2 and 3 for any other views you want to use as a filter.

   Now, selecting marks in these controlling views filters data in one or more detail views—all without affecting any other controlling views.



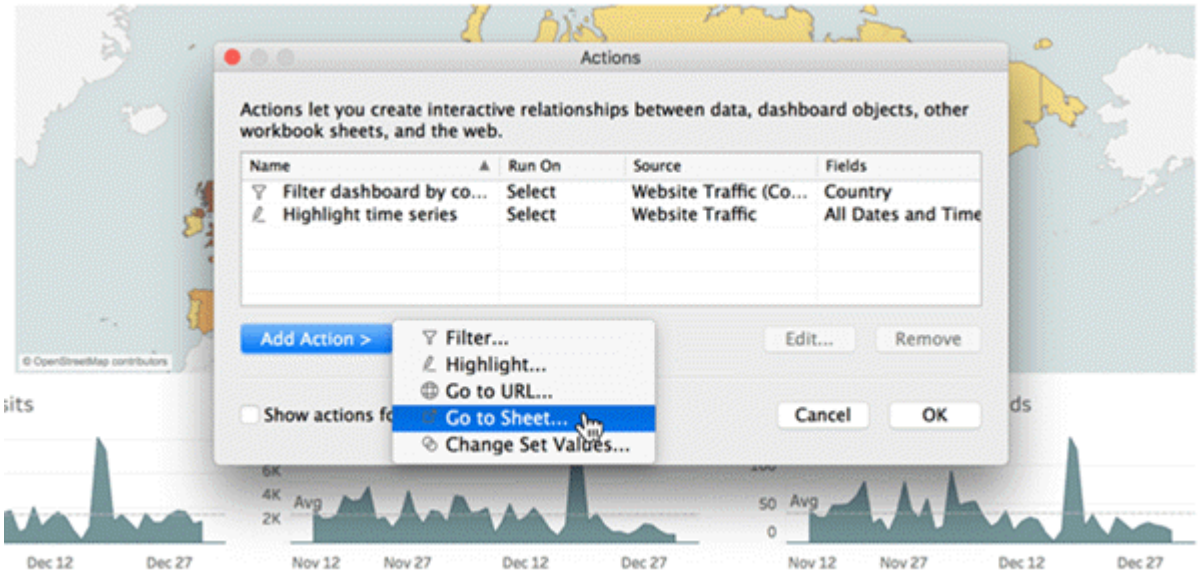
Navigate from One View to Another View, Dashboard, or Story:

The Go to Sheet action allows users to quickly navigate to a related visualization—whether it's a dashboard, sheet, or story—by clicking on a mark or a tooltip menu item in the original view.

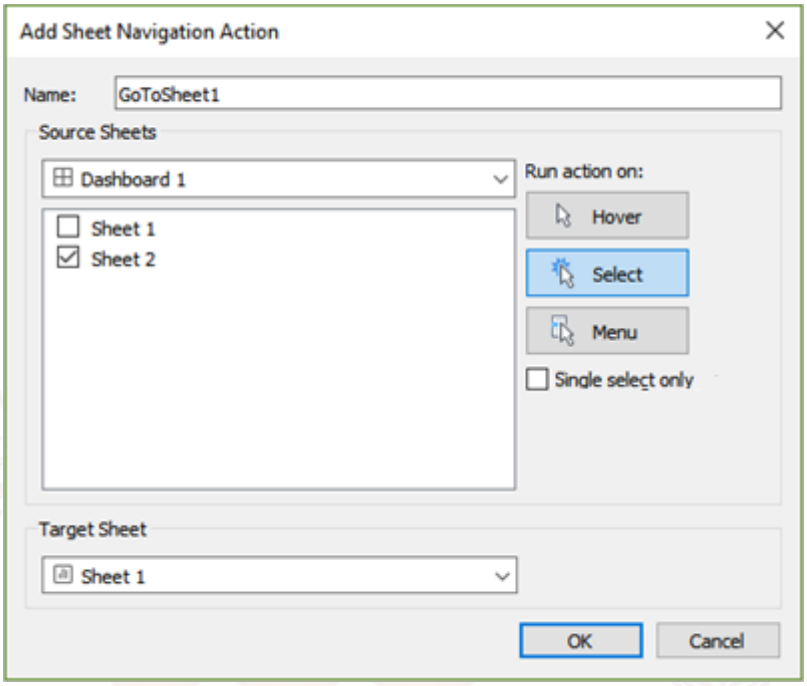
Note: If you want to navigate between dashboards without requiring user interaction with the data, use the Navigation object instead.

Steps:

1. From your dashboard, go to Dashboard > Actions.
2. In the Actions dialog box, click Add Action, then select Go to Sheet.



3. Specify a name for the action. (If you choose to run the action using a tooltip menu, the name you specify here is what the tooltip displays.)



4. Choose a source sheet that will trigger the action.
5. Define how users will activate the navigation action on the dashboard. Options like Select or Menu work best for navigation actions.
   - If you select Select, consider enabling Single-select only to prevent users from navigating away when multiple marks are selected.
6. For the target sheet, specify the destination that will display when users click marks or tooltip menu items in the source sheet. Click OK to finalize

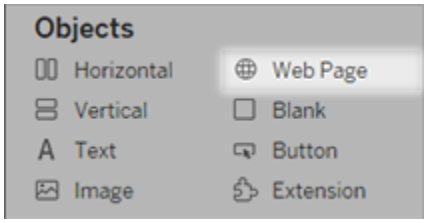
Display a Web Page Interactively in a Dashboard (Using Tableau Desktop):

You can use a URL action with a web page object to display web-based information interactively within a dashboard. For instance, if your dashboard shows profits by country, you can enhance it by displaying supplementary details about the countries from a website.

Tip: Rename web page objects to organize and target them more efficiently in the dashboard.

Steps:

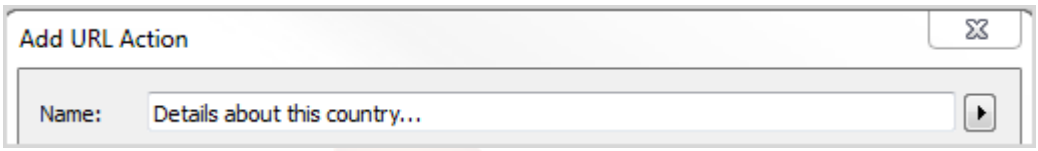
a) Drag a Web Page object onto your dashboard and enter the desired URL.



b) In your dashboard, go to Dashboard > Actions.

c) In the Actions dialog box, click Add Action and select Go to URL.

d) Enter a name for the link. If the action is set to run through a menu, such as a tooltip menu option, this name will appear as the displayed option.

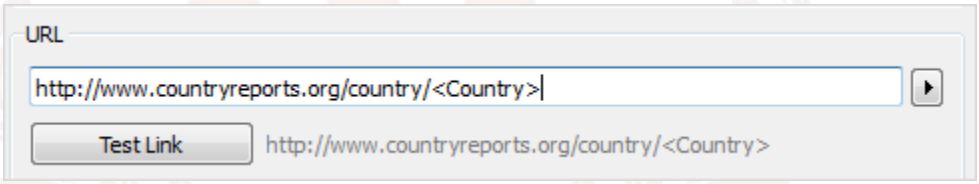


e) Under Source Sheets, choose the view or data source that will trigger the action. For instance, if the action should be activated when a user clicks a link in a map's tooltip, select the map view.

f) Decide how users will activate the action—on hover, select, or via the menu. For more details, refer to Running Actions.

g) Enter the URL, including the http:// or https:// prefix (e.g., http://www.example.com).

- You can incorporate field values as parameters in the URL. For example, if your dashboard uses the Country field, you can include <Country> as a parameter in the URL.



h) For URL Target, select Web Page Object, and select the object you created in step 1.

When you launch the action, a web page automatically loads within the dashboard rather than opening a separate browser window.

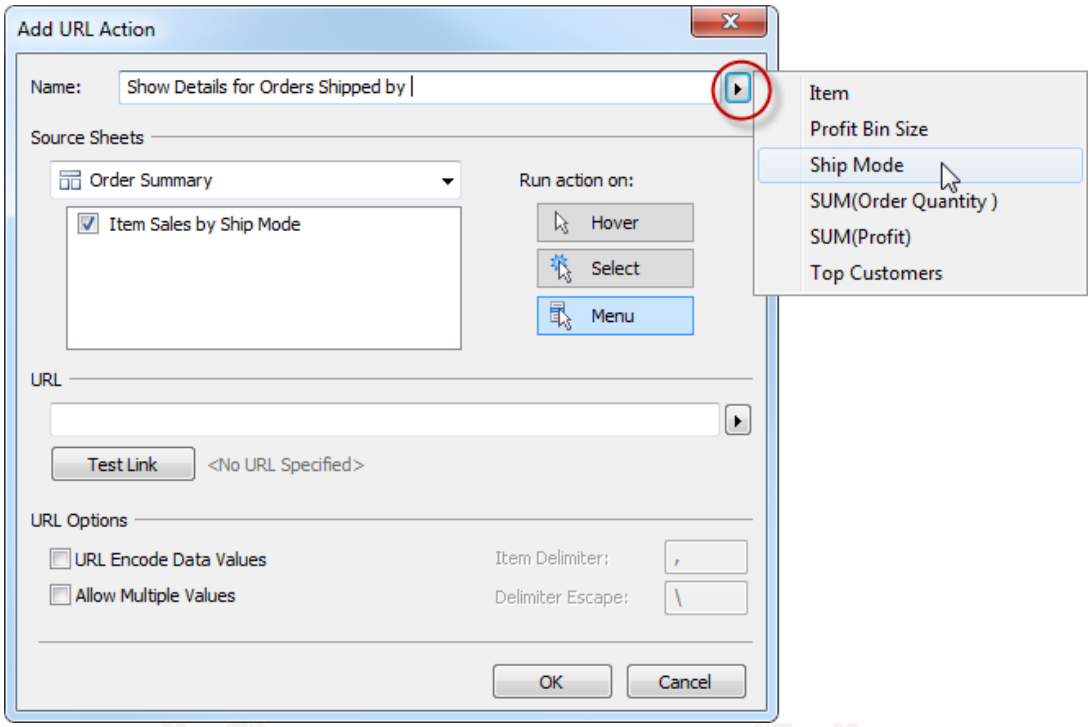


Using Field and Filter Values in Action Names

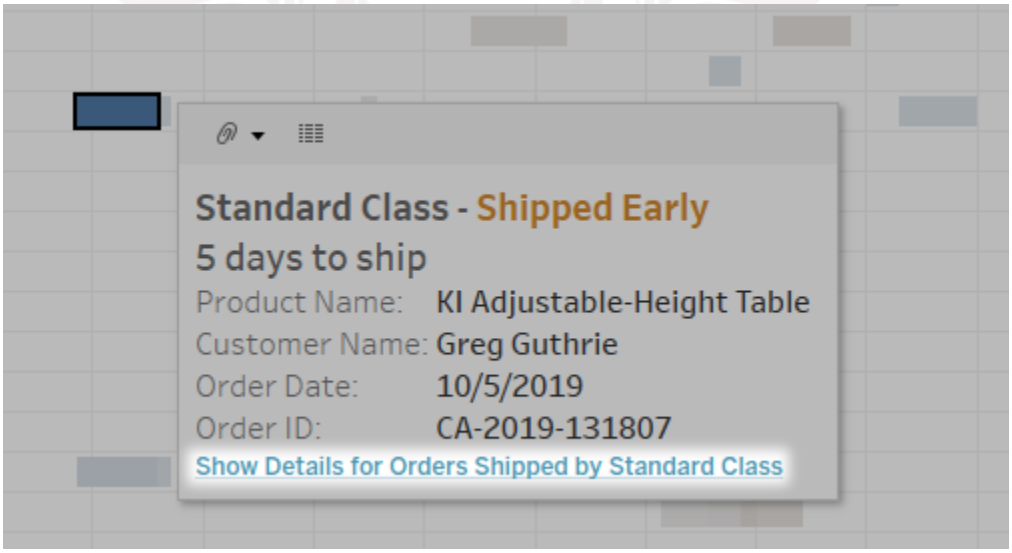
In addition to using field, parameter, and filter values in URLs, you can use this information as variables in the action names. The name of the action displays on the context menu when an action is launched using the menu. Using field and filter variables in the name is useful in making the action specific to the selected mark. In a view showing real estate information, you could name a URL action that points at satellite images from an online mapping service, “Show satellite image of <Address>.” When you right-click (control-click on Mac) a specific mark, the <Address> tag is replaced with the location value associated with that mark.

- In the Add Action dialog box, begin typing the name for the action.

- Place the cursor where you want to insert the field or filter value.
- Click the arrow to the right of the text box and select the field or filter you want to add as a variable. The field or filter name is added between angle brackets.



Here's a finished example for an action run from a tooltip menu:



8. SUMMARY

This chapter focuses on creating effective dashboards that facilitate data exploration and analysis. It begins with an overview of building dashboards for data exploration, emphasizing how to design layouts that enhance usability and provide meaningful insights. By integrating multiple visualizations, dashboards can help users understand complex datasets through a cohesive and interactive interface. The chapter progresses to combining multiple charts into interactive dashboards, showcasing techniques to bring together diverse charts like maps, bar charts, and tables into a single view. This enables users to analyze data from multiple perspectives within the same context, streamlining the decision-making process.

To enhance the user experience, adding interactivity with filters is discussed, highlighting how filters can dynamically adjust the data displayed in various views. This section explains how to apply filters to specific sheets or across an entire dashboard, allowing users to customize their analysis based on their needs. Highlights are introduced as a way to emphasize specific data points while retaining the context of the broader dataset. By enabling tools like highlighters or legend highlighting, users can focus on particular areas of interest and gain deeper insights without losing sight of the overall data trends.

Finally, the chapter delves into dashboard actions, which are powerful tools for creating interactivity. Actions like filter, highlight, and URL actions allow users to navigate between sheets, drill down into detailed data, or access external resources directly from the dashboard. These features make dashboards more engaging and actionable, enhancing their value for business intelligence and analytics.

9. GLOSSARY

Dashboard	-	A visual interface that combines multiple charts, views, and elements like filters, actions, and highlighters to present and analyze data interactively in Tableau.
Interactivity	-	The ability for users to interact with a Tableau dashboard or view, allowing them to filter, highlight, and navigate between data points or sheets for deeper exploration.
Filter	-	A mechanism in Tableau that allows users to refine the data displayed in a dashboard by specifying certain criteria (e.g., time range, category, or region).
Highlighter	-	A graphical tool used to visualize relationships or correlations between two numerical variables, often revealing trends or patterns.
Legend Highlighting	-	A feature in Tableau that highlights data marks corresponding to a specific item in a legend, helping users compare and focus on certain categories in the view.
Actions	-	Interactive features in Tableau dashboards that enable users to filter, highlight, or navigate between different views and external resources. Types of actions include filter, highlight, URL, and Go to Sheet actions.
Go to Sheet Action	-	A Tableau dashboard action that allows users to navigate to another sheet or dashboard by selecting a mark or a menu item.
URL Action	-	A type of dashboard action in Tableau that links to an external URL, such as a webpage or online resource, from a dashboard or view.
Highlight Action	-	A type of Tableau action that emphasizes data points based on a selection or interaction in the dashboard, highlighting matching values in other views.
Filter Action	-	A dashboard action in Tableau that refines the data in other views based on the selection made in the source view.
Web Page Object	-	A dashboard element in Tableau used to display a webpage inside the dashboard, which can be linked to a URL action.

Discrete Field	-	A type of field in Tableau that has distinct, individual values (e.g., categories like product names, regions, or dates) that can be used to filter or highlight data in a view.
Data Storytelling	-	The process of using visualizations and dashboards to present a narrative, providing insights and context through data analysis and interactive elements.
Navigation Object	-	A Tableau element used to guide users between different dashboards or views without requiring direct interaction with data marks, often using buttons or links.
Context Filter	-	A special type of filter in Tableau that is applied first in a sequence of filters, helping to refine data by setting a context for other filters.
Tooltip	-	A small pop-up box that appears when a user hovers over a data point in a visualization, providing additional information about the data point.
Dashboard Layout	-	The arrangement of charts, views, filters, and other interactive elements within a Tableau dashboard, ensuring clarity and ease of use.
Field Values as Parameters	-	The use of dynamic field values as parameters in URLs or actions to personalize the information displayed in a dashboard based on user selections.
Data Exploration	-	The process of interacting with data through filters, highlights, and actions to uncover insights and identify patterns or trends.
Single-Select Only	-	A setting in Tableau that restricts users to selecting only one mark or data point at a time in a view or dashboard, often used to prevent unintended navigation or filtering when multiple items are selected.

10. SELF-ASSESSMENT QUESTIONS

Fill in the Blank:	
1	Combining multiple charts into a single dashboard helps users analyze data from multiple _____ within the same context.
2	Adding interactivity with filters enables users to dynamically adjust the data displayed based on their _____.
3	To emphasize data trends, users can use tools like highlighters or _____ highlighting.
True or False	
4	Filters in dashboards can only be applied to one specific sheet at a time.
5	Dashboard actions allow navigation to external resources like web pages.
6	Interactive dashboards combine multiple charts into a single layout to simplify data storytelling.
Multiple Choice Questions:	
7	What is the primary purpose of building dashboards for data exploration? a) To display static reports b) To facilitate dynamic data analysis and insights c) To replace all spreadsheets d) To create only visual designs
8	Which feature allows users to emphasize specific data points while retaining the context of the entire dataset? a) Filters b) Highlights c) Actions d) Static charts
9	What type of action allows users to navigate between sheets, dashboards, or stories? a) Filter action b) Highlight action c) URL action d) Go to Sheet action

10	What is the first step to add a web page object in a Tableau dashboard? a) Add a URL action from the toolbar b) Drag a Web Page object onto the dashboard c) Select the Source Sheets d) Enter a keyword in the filter

11. TERMINAL QUESTIONS

Short Answer Questions

1. Describe the steps involved in creating a dashboard for data exploration in Tableau. Include how interactivity can enhance the user experience.
2. Explain how combining multiple charts into a single interactive dashboard helps in data analysis. Provide examples of scenarios where this technique is particularly useful.
3. What is the role of filters in dashboards? Discuss the types of filters available in Tableau and how they can be applied across views.
4. Highlight the significance of dashboard actions in Tableau. Describe the different types of actions and their use cases with examples.
5. Discuss the importance of using highlights in a dashboard. Explain how highlighters and legend highlighting improve data interpretation and provide an example of their application.

Long Answer Questions

1. Explain the process of designing an interactive dashboard in Tableau, including the integration of multiple charts, filters, highlights, and dashboard actions. Discuss best practices for ensuring usability and clarity.
2. Discuss the role of interactivity in dashboards. Elaborate on the various interactive features available in Tableau, such as filters, highlighters, legend highlighting, and actions, with examples of their applications in real-world scenarios.
3. Describe how dashboard actions enhance the functionality of Tableau dashboards. Provide a detailed explanation of different action types (filter, highlight, URL, and Go to Sheet) and their use cases. Include examples to illustrate your answer.
4. What are the advantages of combining multiple charts in Tableau dashboards for data storytelling and exploration? Discuss how tools like filters, highlighters, and actions contribute to creating an intuitive and insightful user experience.
5. Evaluate the importance of highlights in Tableau dashboards. Compare the functionalities of highlighters and legend highlighting, and explain how they contribute to effective data interpretation and decision-making with suitable examples.

12. ANSWERS

Self-Assessment Questions

1. perspectives
2. needs
3. legend
4. False
5. True
6. True
7. b
8. b
9. d
10. b

Terminal Questions Answers

Short Question Answers

1. **Steps for Creating Dashboards for Data Exploration:** Start by selecting relevant views or charts and arranging them in a logical layout. Ensure the design is user-friendly and emphasizes key insights. Add interactivity by incorporating filters, highlights, and actions, which allow users to explore data dynamically. Interactivity enhances the experience by enabling custom analyses, drilling into specific details, and maintaining flexibility in how data is viewed.
2. **Combining Multiple Charts for Analysis:** Integrating various charts into a single dashboard helps users analyze data from multiple perspectives simultaneously. For example, combining a map for geographical trends, a bar chart for category-wise breakdowns, and a line chart for trends over time offers a holistic view of the data. This technique is particularly useful in business intelligence scenarios, such as sales performance analysis or customer segmentation.
3. **Role and Types of Filters:** Filters refine the data displayed in dashboards, enabling users to focus on specific subsets. Tableau offers several filter types, including dimension filters, measure filters, and context filters, which can be applied to individual sheets or across all views. Filters improve usability and flexibility, allowing users to tailor their analysis based on criteria like time range, categories, or regions.

4. **Dashboard Actions and Their Significance:** Dashboard actions create interactivity by linking views and enabling user-driven navigation or data exploration. Types of actions include filter actions (refining data across views), highlight actions (emphasizing related data points), and URL actions (linking to external resources). For instance, a filter action can update charts based on a selected region, while a URL action might display related web-based information.

5. **Importance of Highlights in Dashboards:** Highlights emphasize specific data points while retaining overall context, aiding in focused analysis. Highlighters allow keyword searches or selections to dynamically highlight marks, while legend highlighting focuses on marks associated with a particular category. These features are invaluable for identifying patterns or outliers, such as pinpointing high-performing products in a sales dashboard.

Long Question Answers

1. **Designing an Interactive Dashboard in Tableau:**

   Designing an interactive dashboard in Tableau begins by selecting relevant data views such as maps, bar charts, tables, and scatter plots, and arranging them logically to ensure the dashboard communicates key insights effectively. Integration of multiple charts helps users analyze data from different perspectives. Filters are applied to allow users to refine the data they see, focusing on specific categories or time periods. Highlighters are added to emphasize key data points, and legend highlighting can be used to focus on categories. Dashboard actions, such as filter, highlight, and URL actions, make the dashboard interactive by enabling users to navigate between views or external resources. Best practices for designing effective dashboards include maintaining simplicity, using clear labeling, providing context for the data, and ensuring that interactive elements like filters are intuitive and easily accessible.

2. **Role of Interactivity in Dashboards:**

   Interactivity plays a crucial role in making dashboards more dynamic and user-driven, allowing users to customize their views. Tableau offers multiple interactive features:

   - **Filters** allow users to refine the data shown, making it easier to focus on specific subsets, such as particular time periods or categories.
   - **Highlighters** enable users to highlight specific data points for a more detailed exploration.
   - **Legend highlighting** helps to draw attention to marks that are associated with specific values in the legend, simplifying comparisons across categories.
   - **Actions** like **filter** and **highlight actions** can automatically update other views based on user selections, creating a seamless interaction between multiple visualizations. For example, in a

sales dashboard, selecting a region on a map could automatically filter a bar chart to show sales performance for that region. These interactive features allow users to engage deeply with the data, uncover trends, and make more informed decisions.

3. **Dashboard Actions and Their Functionality:**

Dashboard actions in Tableau enhance user interactivity by creating links between different views, making the dashboard dynamic. The different action types are:

- **Filter actions** refine the data shown in other views based on user interaction with one view. For example, selecting a region on a map filters a table to only show data for that region.
- **Highlight actions** emphasize data points that correspond to the selected element in another view, such as highlighting sales numbers for a selected product category.
- **URL actions** allow the dashboard to link to external resources, like websites or online documents, which can supplement the data presented in the dashboard.
- **Go to Sheet actions** navigate users to different sheets or dashboards upon selecting an element. This action is useful for guiding users to more detailed views or related content. These actions make dashboards more interactive, providing users with immediate access to filtered or highlighted information and enabling navigation to more detailed views or external resources.

4. **Combining Multiple Charts for Data Storytelling and Exploration:**

Combining multiple charts into a single Tableau dashboard allows users to explore data from various angles and helps convey a coherent story. For example, a combination of a map, bar chart, and line graph in a dashboard enables users to explore geographical, categorical, and temporal data in a unified view. **Filters** help users refine what they see, such as filtering the map to a specific region, and dynamically updating the bar chart and line graph. **Highlighters** can be used to call attention to specific data points, such as highlighting the top-performing products in the bar chart. **Dashboard actions** enable users to drill down into more detailed data or link to related content, enhancing the user experience. The combination of these features allows for more insightful exploration and better decision-making, as users can easily cross-reference different types of data and dive deeper into specific insights.

5. **Importance of Highlights in Tableau Dashboards:**

Highlights are essential for drawing attention to specific data points in a Tableau dashboard while maintaining context. **Highlighters** let users search for and highlight specific marks based on keywords or field values, helping users focus on particular areas of interest. For example, in a sales

dashboard, users can search for a product or region to quickly highlight relevant data. **Legend highlighting** is used to highlight all data points corresponding to a specific value in the legend, such as highlighting all sales from a particular region or product category. These tools are particularly useful for making comparisons, spotting trends, and identifying outliers or anomalies. For example, legend highlighting can help users visually compare sales figures for different regions, while highlighters can quickly identify a specific product's performance across regions. By providing a clear focus on specific data points, highlights improve the interpretability of the data and assist in decision-making.

13. SUGGESTED BOOKS AND REFERENCES

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